# Dimensionality Reduction With Enhanced Hybrid-Graph Discriminant Learning for Hyperspectral Image Classification

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***Abstract*— Dimensionality reduction (DR) is an important way of improving the classification accuracy of a hyperspectral image (HSI). Graph learning, which can effectively reveal the intrinsic relationships of data, has been widely used in the case of HSIs. However, most of them are based on a simple graph to represent the binary relationships of data. An HSI contains complex high-order relationships among different samples. Therefore, in this article, we propose a hybrid-graph learning method to reveal the complex high-order relationships of the HSI, termed enhanced hybrid-graph discriminant learning (EHGDL). In EHGDL, an intraclass hypergraph and an interclass hypergraph are constructed to analyze the complex multiple relationships of a HSI. Then, a supervised locality graph is applied to reveal the binary relationships of a HSI which can form the complementarity of a hypergraph. Simultaneously, we also construct a weighted neighborhood margin model to boost the difference of samples from different classes. Finally, we design a DR model based on the intraclass hypergraph, the interclass hypergraph, the supervised locality graph, and the weighted neighborhood margin to improve the compactness of the intraclass samples and the separability of the interclass samples, and an optimal projection matrix can be achieved to extract the low-dimensional embedding features of the HSI. To demonstrate the effectiveness of the proposed method, experiments have been conducted on the Indian Pines, PaviaU, and HoustonU data sets. The experimental results show that EHGDL can generate better classification performance compared with some related DR methods. As a result, EHGDL can better reveal the complex intrinsic relationships of a HSI by the complementarity of different characteristics and enhance the discriminant performance of land-cover types.**

***Index Terms*— Hyperspectral image (HIS) classification, dimensionality reduction (DR), graph learning, locality-constrained linear coding (LLC), neighborhood margin.**

## I. Introduction

SPECTRAL imaging is a very effective technique to analyze the composition of each material [1], [2] due to the fact that different materials possess different properties under different electromagnetic spectra [3]. This technique has been widely used in Earth observation to discriminate the objects of the Earth's surface [4], [5]. The captured image is considered as a hyperspectral cube, termed a hyperspectral image (HSI), which contains hundreds of contiguous narrow bands covering the visible-to-infrared spectra [6], [7]. Compared with the traditional optical image, a HSI contains more abundant information which has attracted the researchers' attention in the field of image processing, such as classification [8], clustering [9], unmixing [10], target detection [11], anomaly detection [12], and endmember extraction [13], in which classification is a fundamental task to discriminate the material types of each pixel in the HSI.

For classification, many methods have been generated to automatically judge the category of each pixel in the HSI [14], [15]. Based on different similarity measure strategies, the research studies proposed the $k$-nearest-neighbor (KNN) [16] and spectral angle mapper (SAM) [17] classification methods, which, respectively, use the minimum Euclidean distance and spectral angle between an unknown pixel and each priori pixel to discriminate the class of the unknown pixel. With the probability distribution model, some probability classification methods, including naïve Bayes classifier (NBC) [18], Gaussian mixture model classifier (GMMC) [19], and maximum likelihood classifier (MLC) [20], were proposed by applying the probability of an unknown pixel in each class to obtain its type. They do not generally achieve satisfactory classification results because these classification methods are just based on some simple criteria to construct these classification models. To boost the performance of classification, some advanced methods have been designed with some complex

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decision criteria, such as support vector machine (SVM) [21], extreme learning machine (ELM) [22], random forest (RF) [23], and rotation forest (RoF) [24]. The abovementioned classification methods directly utilize the observed data to achieve classification results. However, the original HSI, with a large number of bands and strong correlation, typically contains high dimensionality and massive redundant information [25], [26], which will result in the curse of dimensionality and affect the performance of classification [27]. In addition, it will cause the Hughes phenomenon to directly classify the original data with these classifiers, that is to say, the classification accuracy will first rise and then decline with the increase of dimensionality under limited samples [28].

To reduce the dimensionality and the redundancy, dimensionality reduction (DR) is an effective technique which includes feature selection and feature transformation. Feature selection is to choose several discriminative bands from the original data to improve the classification accuracy of HSI, while feature transformation is to map the high-dimensional data into a low-dimensional feature space which can extract some desired information to boost the performance of classification [29], [30]. Some classic feature transformation methods have been designed on the basis of statistical theory, such as principal component analysis (PCA) [31], linear discriminant analysis (LDA) [31], maximum noise fraction (MNF) [32], nonparametric weighted feature extraction (NWFE) [33], and maximum margin criterion (MMC) [34]. These statistic-based methods, which assume that the samples obey some statistic distributions, do not consider the intrinsic structure of data and cannot usually achieve satisfactory results for the complex data distribution. To analyze the intrinsic structure of data, the researchers develop the manifold learning methods to reveal the manifold characteristics of high-dimensional data, including isometric mapping (Isomap) [35], locally linear embedding (LLE) [36], and Laplacian eigenmaps (LE) [37]. However, Isomap, LLE, and LE cannot obtain a projection matrix to directly transform the out-of-sample data into the low-dimensional feature space. To address the out-of-sample extension problem, LLE and LE are developed as two linear projection methods—neighborhood preserving embedding (NPE) [38] and locality preserving projections (LPPs) [39], respectively.

To unify these statistic and manifold techniques, graph embedding was proposed to reveal the intrinsic relationships of data based on some statistic or geometric properties [40]. According to this framework, many DR methods can be reconstructed, such as PCA, LDA, LLE, LE, NPE, and LPP. The differences of these methods are the definition of a similarity matrix and constraint matrix. In addition, the novel DR methods can be designed based on graph embedding, including local Fisher discriminant analysis (LFDA) [41], regularized local discriminant embedding (RLDE) [42], and local geometric structure Fisher analysis (LGSFA) [43]. For these methods, they use the Euclidean distance to represent the graph structure of data. However, the neighborhood construction method is easily impacted by data noise and the number of the neighbors is a fixed value that may generate unreasonable neighborhood structure [44].

To overcome the problems of graph construction, some adaptive graph methods were proposed on the basis of sparse representation. The sparse representation uses a dictionary to linearly represent a sample [45]. The obtained representation coefficients are very sparse, that is to say, most of the coefficients are zeros and a few coefficients are nonzeros to reflect the intrinsic relationships of the corresponding samples [46]. Based on the sparse coefficients, the intrinsic relationships can be adaptively represented and an adaptive graph can be constructed. The sparse graph is robust to data noise and possesses datum-adaptive structures for each sample [44]. With the sparse graph, many sparse representation methods have been proposed to achieve the DR of data, and some common methods based on sparse graph include sparsity preserving projection (SPP) [47], sparsity preserving graph embedding (SPGE) [48], sparse neighborhood preserving embedding (Sparse NPE) [44], and sparsity preserving analysis (SPA) [49], in fact, SPGE and sparse NPE are identical to SPP. But, they are unsupervised sparse graph embedding methods and cannot generally generate effective discriminating features in some cases. To improve the effect of DR, some supervised methods were proposed on the basis of the sparse graph, such as discriminant sparsity NPE (DSNPE) [50], discriminative learning by sparse representation projection (DLSP) [51], sparse graph-based discriminant analysis (SGDA) [48], and sparse discriminant embedding (SDE) [52]. The sparse graph methods tend to individually represent the locally linear structure of each sample, which neglects the global properties of data. Therefore, the low-rank graph-based discriminant analysis (LGDA) [53] method was developed to reveal the global structure. To consider the local and global information, the researchers simultaneously use the sparse and low-rank properties of data to design the sparse and low-rank graph-based discriminant analysis (SLGDA) [53] algorithm. In addition, among the samples are competitive in sparse representation, while some researchers discovered that among the samples are collaborative, and they proposed a collaborative graph-based discriminant analysis (CGDA) [54] method.

The abovementioned graph learning methods just consider binary relationships of data that only two samples are connected with an edge. In real application, many data generally possess complex high-order relationships, that is to say, each class is relevant to more than two samples [55]. To represent the complex intrinsic relationships of data, researchers introduced the hypergraph theory into the field of machine learning and constructed many hypergraph learning algorithms. A hypergraph contains vertices and hyperedges with weights, and each hyperedge has multiple vertices (generally more than two vertices) while a simple graph just possesses two vertices on each edge [56]. Recently, hypergraph learning has been widely used in the fields of image segmentation, clustering, classification, and embedding [55]–[57]. For DR, many methods have been designed based on hypergraph including discriminant hyper-Laplacian projection (DHLP) [57], binary hypergraph (BH) [58], spatial hypergraph (SH) [58], hypergraph embedding-based spatial–spectral joint features (SSHG) [59], and semi-supervised hypergraph embedding (SHGE)

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[60]. In addition, the hypergraph model also achieved good performance for the band selection [61], unmixing [62], and classification [63] of the HSI.

These hypergraph learning methods mainly analyze the high-order relationships to represent the homogeneity of all the related samples and weaken the binary relationships that reflect the similar property of two samples [64], [65]. However, different samples have different spectral features in the HSI [66]. To enhance the homogeneity of the related samples and reduce the heterogeneity of the two same class samples for the complex HSI, we simultaneously consider the high-order relationships and the binary relationships to design an enhanced hybrid-graph discriminant learning (EHGDL) method for the DR of HSI. The EHGDL algorithm applies two discriminant hypergraphs (revealing the high-order relationships), a supervised locality graph (revealing the binary relationships), and a weighted neighborhood margin (revealing the margin information) to construct a feature embedding model. EHGDL considers not only the multiple properties but also the binary characteristics. Simultaneously, it also analyzes the margins of different classes. Therefore, EHGDL is very effective to reveal the common features of samples from the same class and represent the difference of samples from different classes. The main contributions of this article include the following.

1) According to the idea of a hypergraph, we propose two discriminant hypergraphs, i.e. an intraclass hypergraph and an interclass hypergraph, based on the intraclass and interclass neighbors, to represent the high-order relationships of HSI, which is beneficial to analyze the complex multiple structures and reveal intrinsic properties of HSI.
2) To reveal the binary relationships of HSI, a supervised locality graph is constructed with label information and locality-constrained coding. The binary graph can be considered as the complementarity of two discriminant hypergraphs to obtain the other structural information of the HSI, and then the binary characteristics and the multiple structures will form the complementary features to better reflect the intrinsic properties of the HSI.
3) To enhance the difference of samples from different classes, we apply the Kullback–Leibler divergence (KLD) to design a weighted neighborhood margin for separating the similar samples from different classes as far as possible. The weighted neighborhood margin will improve the discriminant performance of different land-cover types, especially, for the precise classification of the similar land-cover (e.g., stressed grass and synthetic grass).
4) In addition, we also propose a new DR model to compact the intraclass properties from the intraclass hypergraph and the supervised locality graph and to separate the interclass characteristics from the interclass hypergraph and the weighted neighborhood margin. Then, we can achieve an optimal projection matrix to extract the discriminant features of HSI and three public data sets are used to show the performance of the proposed method.

The rest of this article is organized as follows. Section II briefly reviews some related works including graph, hypergraph, MMC, and locality-constrained linear coding (LLC). Section III explains the proposed EHGDL method. Experimental results are displayed in Section IV to demonstrate the performance of EHGDL. Finally, Section V gives some conclusions and suggestions.

## II. Related Works

For a HSI, we use a vector $\mathbf{x}_i \in \Re^D$ $(i = 1, 2, \ldots, n)$ to represent a pixel, where $n$ and $D$ represent the number of pixels and bands, respectively. $\ell(\mathbf{x}_i) \in \{1, 2, \ldots, c\}$ denotes the class label of $\mathbf{x}_i$, where $c$ is the class number of land-cover. $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n] \in \Re^{D\times n}$ is a HSI data set. $\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_n] \in \Re^{d\times n}$ is the low-dimensional features of $\mathbf{X}$, where $d$ represents the embedding dimension. With a projection matrix $\mathbf{V} \in \Re^{D\times d}$, the low-dimension features can be represented as $\mathbf{Y} = \mathbf{V}^T\mathbf{X}$.

### A. Graph Learning

A graph is very effective to represent the relationship between the paired samples, which can reveal some of the similar statistical or geometrical properties of data. For an undirected graph $\mathbf{G}$, it can be represented as $\mathbf{G} = \{\mathbf{X}, \mathbf{E}, \mathbf{W}\}$, where $\mathbf{X}$ is the vertex set, $\mathbf{E}$ is the edge set, and $\mathbf{W} = [w_{ij}]_{i,j=1}^n$ is the weight matrix of edges. To construct a graph, some similarity measure methods are adopted to define the connected edge between two vertices and their corresponding weight. If vertices $i$ and $j$ are similar, an edge should be given between vertices $i$ and $j$, and a similar weight of the edge should be defined at the same time.

In low-dimensional space, the graph model should be preserved and similar samples should be compacted as much as possible. Therefore, the objective function can be represented as the following:

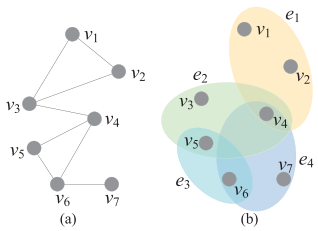
$$\min_{\mathbf{Y}} \frac{1}{2}\sum_{i\neq j}\|\mathbf{y}_i - \mathbf{y}_j\|^2 w_{ij} = \min_{\mathbf{Y}} \operatorname{tr}(\mathbf{YLY}^T)$$
$$\text{s.t. } \operatorname{tr}(\mathbf{YHY}^T) = h \tag{1}$$

where $h$ is a constant, $\mathbf{H}$ is a constraint matrix which is used to avoid a trivial solution of the objective function, $\mathbf{L} = \mathbf{D} - \mathbf{W}$ is the Laplacian matrix of graph $\mathbf{G}$, and $\mathbf{D} = \operatorname{diag}([\sum_{j=1}^n w_{1j}, \sum_{j=1}^n w_{2j}, \ldots, \sum_{j=1}^n w_{nj}])$ is a diagonal matrix. In general, $\mathbf{H}$ is an identity matrix for scale normalization or a Laplacian matrix of penalty graph that is used to suppress some of the unwanted properties of data.

### B. Hypergraph Learning

For hypergraph and graph, the main difference is the number of vertex on each edge. In graph, each edge just has two vertices, while each edge can contain more than two vertices in hypergraph. For a hypergraph, it is defined as $\mathbf{G}^H = \{\mathbf{X}, \mathbf{E}^H, \mathbf{W}^H\}$, where $\mathbf{X}$ is the vertex set, $\mathbf{E}^H$ is the hyperedge set, $\mathbf{W}^H$ is the weight matrix of hyperedges, and each hyperedge $e_i \in \mathbf{E}^H$ has a weight $w_i^e$. An incidence matrix $\mathbf{H} = [h_{ij}^{ve}]_{i,j} \in \Re^{|\mathbf{V}^H|\times|\mathbf{E}^H|}$ is used to represent the relationship

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| | $e_1$ | $e_2$ | $e_3$ | $e_4$ |
|---|---|---|---|---|
| $v_1$ | 1 | 0 | 0 | 0 |
| $v_2$ | 1 | 0 | 0 | 0 |
| $v_3$ | 0 | 1 | 0 | 0 |
| $v_4$ | 1 | 1 | 0 | 1 |
| $v_5$ | 0 | 1 | 1 | 0 |
| $v_6$ | 0 | 0 | 1 | 1 |
| $v_7$ | 0 | 0 | 0 | 1 |

(c)

Fig. 1. Comparison of two graphs. (a) Binary graph $\mathbf{G}$. (b) Hypergraph $\mathbf{G}^H$. (c) Incidence matrix $\mathbf{H}$.

between vertex $v_i$ and hyperedge $e_j$, and $h_{ij}^{ve}$ is defined as follows:

$$h_{ij}^{ve} = \begin{cases} 1, & \text{if } v_i \in e_j \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

To reveal the properties of hypergraph, the degree of vertex $v_i$ is the summation of the hyperedge weights via vertex $v_i$, and the degree of hyperedge $e_i$ is the number of vertices on $e_i$. Therefore, the degrees of $v_i$ and $e_j$ are defined as

$$d_i^v = \sum_{j=1}^{n} w_j^e h_{ij}^{ve} \tag{3}$$

$$d_j^e = |e_j| = \sum_{i=1}^{n} h_{ij}^{ve}. \tag{4}$$

The comparison of graph and hypergraph is shown in Fig. 1. According to Fig. 1(a), each edge of the graph contains only two vertices such as $v_1$ and $v_2$. In Fig. 1(b), this hypergraph has seven vertices and four hyperedges to reveal the complex high-order relationships, and each hyperedge has more than two vertices such as $v_1$, $v_2$, and $v_4$ belonging to $e_1$.

In low-dimensional space, the samples from the same hyperedge should be compacted as close as possible, and the objective function can be represented as the following:

$$\min_{\mathbf{Y}} \frac{1}{2} \sum_{e_i \in \mathbf{E}^H} \frac{w_i^e}{d_i^e} \sum_{(\mathbf{x}_j, \mathbf{x}_k) \in e_i} \left\| \frac{\mathbf{y}_j}{\sqrt{d_j^v}} - \frac{\mathbf{y}_k}{\sqrt{d_k^v}} \right\|^2 = \min_{\mathbf{Y}} \operatorname{tr}(\mathbf{Y}\mathbf{L}^H\mathbf{Y}^T)$$
$$\text{s.t. } \operatorname{tr}\left(\frac{1}{n}\mathbf{Y}\mathbf{Y}^T\right) = 1 \tag{5}$$

where $\mathbf{L}^H = \mathbf{I} - (\mathbf{D}^v)^{-1/2}\mathbf{H}\mathbf{W}^H(\mathbf{D}^e)^{-1}(\mathbf{H})^T(\mathbf{D}^v)^{-1/2}$ is the hyper-Laplacian matrix. $\mathbf{W}^H = \operatorname{diag}([w_1^e, w_2^e, \ldots, w_n^e])$ is the weight matrix. $\mathbf{D}^e = \operatorname{diag}([d_1^e, d_2^e, \ldots, d_n^e])$ and $\mathbf{D}^v = \operatorname{diag}([d_1^v, d_2^v, \ldots, d_n^v])$ are the diagonal matrices of the vertex and hyperedge degrees.

### C. MMC

The MMC method maps the high-dimensional features into the low-dimensional space with the criterion that maximizes the margin between the intraclass and interclass samples [34]. The between-class and within-class scatter matrices are used to represent the marginal information as the following:

$$\mathbf{S}_b = \sum_{i=1}^{c} n_i (\mathbf{u}_i - \mathbf{u})^T (\mathbf{u}_i - \mathbf{u}) \tag{6}$$

$$\mathbf{S}_w = \sum_{i=1}^{c} \sum_{j=1}^{n_i} \left(\mathbf{x}_j^i - \mathbf{u}_i\right)^T \left(\mathbf{x}_j^i - \mathbf{u}_i\right) \tag{7}$$

where $n_i$ is the number of the $i$th class. $\mathbf{x}_j^i$ is the $j$th sample from the $i$th class. $\mathbf{u} = 1/n \sum_{i=1}^{n} \mathbf{x}_i$ is the mean of $\mathbf{X}$. $\mathbf{u}_i = 1/n_i \sum_{j=1}^{n_i} \mathbf{x}_j^i$ is the mean of the $i$th class.

In low-dimensional space, the maximum margin is defined as the following to separate the samples from different classes:

$$\max_{\mathbf{V}} \operatorname{tr}[\mathbf{V}^T(\mathbf{S}^b - \mathbf{S}^w)\mathbf{V}]. \tag{8}$$

### D. LLC

To achieve good sparse coding performance, the coding model should achieve similar codes for similar samples. For the sparse representation of data, locality is more important than sparsity, since locality must result in sparsity. In LLC, it incorporates locality constraint instead of the sparsity constraint and utilizes the locality constraint to project each sample into its local-coordinate system [67]. Therefore, LLC utilizes the following function to generate coding coefficients:

$$\min_{\mathbf{c}_i} \sum_{i=1}^{n} ||\mathbf{x}_i - \mathbf{B}\mathbf{c}_i||^2 + \lambda ||\mathbf{d}_i \odot \mathbf{c}_i||^2$$
$$\text{s.t. } \mathbf{1}^T \mathbf{c}_i = 1 \tag{9}$$

where $\odot$ is the element-wise multiplication. $\mathbf{B} = [\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_m]$ is an overcomplete dictionary and $m$ denotes the size of dictionary. $\mathbf{d}_i$ is the locality adaptor to control the values of coding coefficients $\mathbf{c}_i$. If two samples are very far away, the corresponding $\mathbf{d}_i$ will be a large value and a small value will be assigned to the corresponding coding coefficients, vice versa. The coding coefficients reveal the similarity between $\mathbf{x}_i$ and $\mathbf{B}$. For $\mathbf{d}_i$, it is defined as

$$\mathbf{d}_i = \exp\left(\frac{\operatorname{dist}(\mathbf{x}_i, \mathbf{B})}{\sigma}\right) \tag{10}$$

where $\operatorname{dist}(\mathbf{x}_i, \mathbf{B}) = [\operatorname{dist}(\mathbf{x}_i, \mathbf{b}_1), \ldots, \operatorname{dist}(\mathbf{x}_i, \mathbf{b}_m)]^T$, and $\operatorname{dist}(\mathbf{x}_i, \mathbf{b}_j)$ is the Euclidean distance between $\mathbf{x}_i$ and $\mathbf{b}_j$. $\sigma$ is used to adjust the weight decay speed.

## III. EHGDL

In this section, we describe the proposed EHGDL method to analyze the complex intrinsic relationships of the HSI and extract the effective low-dimensional discriminant features. First, we construct two discriminant hypergraphs, including an intraclass hypergraph and an interclass hypergraph, based on the intraclass and interclass neighbors to reveal the high-order structures of HSI. Then, a supervised locality graph is constructed with the class label and the locality-constrained coding to represent the binary relationships of the HSI. In addition, to further enhance the difference of the samples from different classes, a weighted neighborhood margin is designed with the KLD neighbors to represent the marginal

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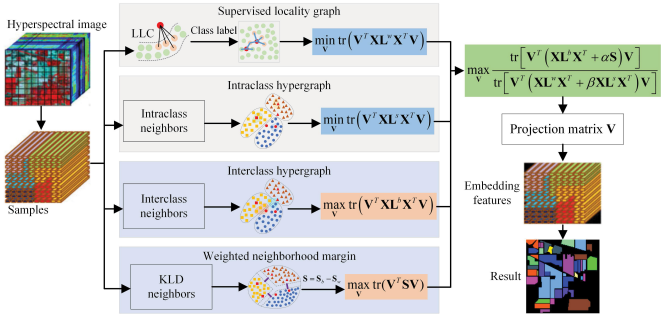



Fig. 2. Flowchart of the proposed EHGDL method.

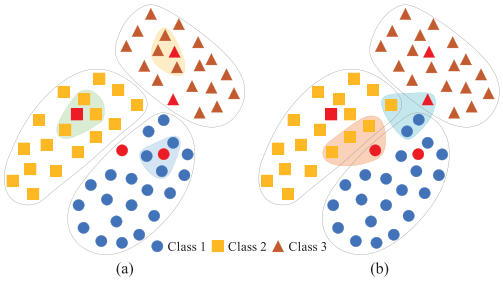



Fig. 3. Construction of (a) intraclass hypergraph and (b) interclass hypergraph.

structures. Finally, a feature embedding model is constructed to achieve projection matrix, and this model not only compacts the properties of the intraclass hypergraph and the supervised locality graph but also separates the characteristics of the interclass hypergraph and the weighted neighborhood margin. Fig. 2 shows the flowchart of EHGDL.

### A. Discriminant Hypergraph

According to the class label of data, two discriminant hypergraphs, including the intraclass and interclass hypergraphs, are constructed to represent the high-order relationships of HSI. We use the intraclass neighbors to construct an intraclass hypergraph $\mathbf{G}^{H_w} = \{\mathbf{X}, \mathbf{E}^{H_w}, \mathbf{W}^w\}$ representing the intraclass multiple relationships, where $\mathbf{X}$, $\mathbf{E}^{H_w}$ and $\mathbf{W}^w$ are the vertices, the intraclass hyperedges, and the intraclass hyperedge weights of the intraclass hypergraph, respectively. The interclass hypergraph $\mathbf{G}^{H_b} = \{\mathbf{X}, \mathbf{E}^{H_b}, \mathbf{W}^b\}$ is designed with the interclass neighbors to reveal the interclass high-order properties of data, where $\mathbf{X}$, $\mathbf{E}^{H_b}$ and $\mathbf{W}^b$ are the vertices, the interclass hyperedges, and the interclass hyperedge weights of the interclass hypergraph, respectively. Fig. 3 is an example to explain the intraclass and interclass hypergraphs.

*1) Intraclass Hypergraph:* After selecting $k_w$ intraclass neighbors using the Euclidean distance, the intraclass incidence matrix $\mathbf{H}^w = [h^w_{i,j}]_{i,j}$ is defined as

$$h^w_{ij} = \begin{cases} 1, & \text{if } \mathbf{x}_j \in \mathcal{N}_{k_w}(\mathbf{x}_i) \text{ and } \ell(\mathbf{x}_i) = \ell(\mathbf{x}_j) \\ 0, & \text{otherwise} \end{cases} \tag{11}$$

where $\mathcal{N}_{k_w}(\mathbf{x}_i)$ is $k_w$ nearest-neighbors (NNs) of $\mathbf{x}_i$ belonging to the same class, and each intraclass hyperedge $e^w_i \in \mathbf{E}^{H_w}$ contains $k_w$ vertices from the same class. For an intraclass hyperedge $e^w_i$, its weight is set as the following to reveal the similarity of these vertices on the intraclass hyperedge:

$$w^w_i = \sum_{\mathbf{x}_j \in \mathcal{N}_{k_w}(\mathbf{x}_i)} \exp\left(-\frac{||\mathbf{x}_j - \mathbf{x}_i||^2}{2\left(t^w_i\right)^2}\right) \tag{12}$$

where $t^w_i = (1/k_w)\sum_{\mathbf{x}_j \in \mathcal{N}_{k_w}(\mathbf{x}_i)} ||\mathbf{x}_j - \mathbf{x}_i||$ is the kernel parameter, and $\mathbf{W}^w = \text{diag}([w^w_1, w^w_2, \ldots, w^w_n])$ is the intraclass weight matrix.

With the intraclass incidence matrix and the intraclass weights, the degrees of $\mathbf{x}_i$ and $e^w_j$ can be represented as

$$d^{vw}_i = \sum_{j=1}^{n} w^w_j h^w_{ij} \tag{13}$$

$$d^{ew}_j = \sum_{i=1}^{n} h^w_{ij}. \tag{14}$$

In low-dimensional embedding space, we should compact the intraclass samples as close as possible. Therefore, the samples on the same intraclass hyperedge should be close in low-dimensional embedding space, and we can get

$$\min_{\mathbf{V}} \frac{1}{2} \sum_{e^w_i \in \mathbf{E}^{H_w}} \frac{w^w_i}{d^{ew}_i} \sum_{(\mathbf{x}_j, \mathbf{x}_k) \in e^w_i} \left\| \frac{\mathbf{V}^T \mathbf{x}_j}{\sqrt{d^{vw}_j}} - \frac{\mathbf{V}^T \mathbf{x}_k}{\sqrt{d^{vw}_k}} \right\|^2. \tag{15}$$

With some mathematical operations, (15) can be further simplified as shown in (16), bottom of the next page, where $\mathbf{L}^w = \mathbf{I} - (\mathbf{D}^{vw})^{-1/2}\mathbf{H}^w\mathbf{W}^w(\mathbf{D}^{ew})^{-1}(\mathbf{H}^w)^T(\mathbf{D}^{vw})^{-1/2}$ represents the intraclass hyper-Laplacian matrix, $\mathbf{D}^{ew} = \text{diag}([d^{ew}_1, d^{ew}_2, \ldots, d^{ew}_n])$ and $\mathbf{D}^{vw} = \text{diag}([d^{vw}_1, d^{vw}_2, \ldots, d^{vw}_n])$ are the intraclass hyperedge and vertex degree matrices.

Therefore, the objective function for the compactness of intraclass data can be denoted as the following:

$$\min_{\mathbf{V}} \ \text{tr}(\mathbf{V}^T \mathbf{X} \mathbf{L}^w \mathbf{X}^T \mathbf{V}). \tag{17}$$

*2) Interclass Hypergraph:* After obtaining the $k_b$ interclass neighbors using Euclidean distance, the incidence matrix $\mathbf{H}^b = [h^b_{i,j}]_{i,j}$ is given by

$$h^b_{ij} = \begin{cases} 1, & \text{if } \mathbf{x}_j \in \mathcal{N}_{k_b}(\mathbf{x}_i) \text{ and } \ell(\mathbf{x}_i) \neq \ell(\mathbf{x}_j) \\ 0, & \text{otherwise} \end{cases} \tag{18}$$

where $\mathcal{N}_{k_b}(\mathbf{x}_i)$ denotes $k_b$ NNs of $\mathbf{x}_i$ from different classes, and $e^b_i \in \mathbf{E}^{H_b}$ is the interclass hyperedge that possesses $k_b$ vertices from other classes.

For an interclass hyperedge $e^b_i$, its weight is used to represent the heterogeneity properties of these vertices on the interclass hyperedge, and it is defined as follows:

$$w^b_i = \sum_{\mathbf{x}_j \in \mathcal{N}_{k_b}(\mathbf{x}_i)} \exp\left(-\frac{||\mathbf{x}_j - \mathbf{x}_i||^2}{2(t^b_i)^2}\right) \tag{19}$$

where $t^b_i = (1/k_b)\sum_{\mathbf{x}_j \in \mathcal{N}_{k_b}(\mathbf{x}_i)} ||\mathbf{x}_j - \mathbf{x}_i||$ is a kernel parameter, and $\mathbf{W}^b = \text{diag}([w^b_1, w^b_2, \ldots, w^b_n])$ is the interclass weight matrix.

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According to the interclass incidence matrix and the interclass weights, we can achieve the degrees of $\mathbf{x}_i$ and $e_j^b$ as follows:

$$d_i^{vb} = \sum_{j=1}^{n} w_j^b h_{ij}^b \tag{20}$$

$$d_j^{eb} = \sum_{i=1}^{n} h_{ij}^b. \tag{21}$$

To better discriminate the interclass samples in low-dimensional embedding space, we should enhance the separability of the interclass samples. Therefore, in low-dimensional space, the samples from different classes should be separated as far as possible, and we design a function to separate the samples on the interclass hyperedge as the following:

$$\max_{\mathbf{V}} \frac{1}{2} \sum_{e_i^b \in \mathbf{E}^{H_b}} \frac{w_i^b}{d_i^{eb}} \sum_{(\mathbf{x}_j, \mathbf{x}_k) \in e_i^b} \left\| \frac{\mathbf{V}^T \mathbf{x}_j}{\sqrt{d_j^{vb}}} - \frac{\mathbf{V}^T \mathbf{x}_k}{\sqrt{d_k^{vb}}} \right\|^2. \tag{22}$$

According to the simplification process of (16), we can obtain the objective function to separate the intraclass data as the following:

$$\max_{\mathbf{V}} \ \mathrm{tr}(\mathbf{V}^T \mathbf{X} \mathbf{L}^b \mathbf{X}^T \mathbf{V}) \tag{23}$$

where $\mathbf{L}^b = \mathbf{I} - (\mathbf{D}^{vb})^{-1/2} \mathbf{H}^b \mathbf{W}^b (\mathbf{D}^{eb})^{-1} (\mathbf{H}^b)^T (\mathbf{D}^{vb})^{-1/2}$ represents the interclass hyper-Laplacian matrix, $\mathbf{D}^{eb} = \mathrm{diag}([d_1^{eb}, d_2^{eb}, \ldots, d_n^{eb}])$ and $\mathbf{D}^{vb} = \mathrm{diag}([d_1^{vb}, d_2^{vb}, \ldots, d_n^{vb}])$ are the interclass hyperedge and vertex degree matrices.

### B. Supervised Locality Graph

According to the locality-constrained coding, the overcomplete dictionary $\mathbf{B}$ is set to $\mathbf{X}$. Then, we can obtain the coding coefficients of $\mathbf{x}_i$ as $\mathbf{c}_i = [c_{i,1}, c_{i,2}, \ldots, c_{i,i-1}, 0, c_{i,i+1}, \ldots, c_{i,n}]$, which can adaptively represent the similar binary relationships of data. For the coding coefficients, only very few of them have large values, and the other coding coefficients are close to zero. Therefore, we just maintain the coding coefficients possessing large values and set the small ones to zero. Then, we use all the coding coefficients and the class label to construct a supervised locality graph $\mathbf{G}^s = \{\mathbf{X}, \mathbf{E}^s, \mathbf{W}^s\}$, where $\mathbf{X}$ is the vertex, $\mathbf{E}^s$ is the edge, and $\mathbf{W}^s$ is the weights of edge. The construction of the supervised locality graph is shown in Fig. 4.

With the coding coefficients, we connect two vertices with an edge if the corresponding coefficient is nonzero, otherwise there is no edge between the vertices. Then, the weight of each edge is set to represent the similarity of two samples, and it is defined as the following:

$$w_{ij}^s = \begin{cases} \theta \dfrac{c_{i,j}}{d_{ij}}, & \text{if } c_{i,j} \neq 0 \text{ and } \ell(\mathbf{x}_i) = \ell(\mathbf{x}_j) \\ -\dfrac{c_{i,j}}{d_{ij}}, & \text{if } c_{i,j} \neq 0 \text{ and } \ell(\mathbf{x}_i) \neq \ell(\mathbf{x}_j) \\ 0, & \text{otherwise} \end{cases} \tag{24}$$

where $\theta \geq 1$ is an intraclass weighted parameter to enhance the contribution and compactness of the samples from the same class, because the intraclass samples are more beneficial to reveal the intrinsic property of each class than the

$$\begin{aligned}
&\frac{1}{2} \sum_{e_i^w \in \mathbf{E}^{H_w}} \frac{w_i^w}{d_i^{ew}} \sum_{(\mathbf{x}_j, \mathbf{x}_k) \in e_i^w} \left\| \frac{\mathbf{V}^T \mathbf{x}_j}{\sqrt{d_j^{vw}}} - \frac{\mathbf{V}^T \mathbf{x}_k}{\sqrt{d_k^{vw}}} \right\|^2 \\
&= \sum_{e_i^w \in \mathbf{E}^{H_w}} \frac{w_i^w}{d_i^{ew}} \sum_{(\mathbf{x}_j, \mathbf{x}_k) \in e_i^w} \mathrm{tr} \left( \frac{\mathbf{V}^T \mathbf{x}_j \mathbf{x}_j^T \mathbf{V}}{d_j^{vw}} - \frac{\mathbf{V}^T \mathbf{x}_j \mathbf{x}_k^T \mathbf{V}}{\sqrt{d_j^{vw} d_k^{vw}}} \right) \\
&= \sum_{e_i^w \in \mathbf{E}^{H_w}} \frac{w_i^w}{d_i^{ew}} \sum_{\mathbf{x}_j, \mathbf{x}_k \in \mathbf{X}} h_{ij}^w h_{ik}^w \mathrm{tr} \left( \frac{\mathbf{V}^T \mathbf{x}_j \mathbf{x}_j^T \mathbf{V}}{d_j^{vw}} - \frac{\mathbf{V}^T \mathbf{x}_j \mathbf{x}_k^T \mathbf{V}}{\sqrt{d_j^{vw} d_k^{vw}}} \right) \\
&= \sum_{e_i^w \in \mathbf{E}^{H_w}} \sum_{\mathbf{x}_j, \mathbf{x}_k \in \mathbf{X}} \left( \frac{h_{ik}^w}{d_i^{ew}} \times \frac{w_i^w h_{ij}^w \mathrm{tr}(\mathbf{V}^T \mathbf{x}_j \mathbf{x}_j^T \mathbf{V})}{d_j^{vw}} - \frac{\mathrm{tr}(\mathbf{V}^T \mathbf{x}_j h_{ij}^w w_i^w h_{ik}^w \mathbf{x}_k^T \mathbf{V})}{d_i^{ew} \sqrt{d_j^{vw} d_k^{vw}}} \right) \\
&= \sum_{\mathbf{x}_j \in \mathbf{X}} \left( \mathrm{tr}(\mathbf{V}^T \mathbf{x}_j \mathbf{x}_j^T \mathbf{V}) \frac{1}{d_j^{vw}} \sum_{e_i^w \in \mathbf{E}^{H_w}} \left( w_i^w h_{ij}^w \frac{\sum_{\mathbf{x}_k \in \mathbf{X}} h_{ik}^w}{d_i^{ew}} \right) \right) \\
&\quad - \sum_{e_i^w \in \mathbf{E}^{H_w}} \sum_{\mathbf{x}_j, \mathbf{x}_k \in \mathbf{X}} \mathrm{tr} \left( \mathbf{V}^T \mathbf{x}_j \frac{1}{\sqrt{d_j^{vw}}} h_{ij}^w w_i^w \frac{1}{d_i^{ew}} h_{ik}^w \frac{1}{\sqrt{d_k^{vw}}} \mathbf{x}_k^T \mathbf{V} \right) \\
&= \mathrm{tr}(\mathbf{V}^T \mathbf{X} \mathbf{X}^T \mathbf{V}) - \mathrm{tr}(\mathbf{V}^T \mathbf{X} (\mathbf{D}^{vw})^{-1/2} \mathbf{H}^w \mathbf{W}^w (\mathbf{D}^{ew})^{-1} (\mathbf{H}^w)^T (\mathbf{D}^{vw})^{-1/2} \mathbf{X}^T \mathbf{V}) \\
&= \mathrm{tr}\{\mathbf{V}^T \mathbf{X} [\mathbf{I} - (\mathbf{D}^{vw})^{-1/2} \mathbf{H}^w \mathbf{W}^w (\mathbf{D}^{ew})^{-1} (\mathbf{H}^w)^T (\mathbf{D}^{vw})^{-1/2}] \mathbf{X}^T \mathbf{V}\} \\
&= \mathrm{tr}(\mathbf{V}^T \mathbf{X} \mathbf{L}^w \mathbf{X}^T \mathbf{V})
\end{aligned} \tag{16}$$

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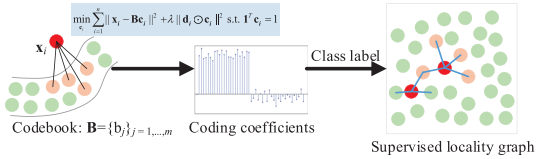



Fig. 4. Construction of the supervised locality graph.

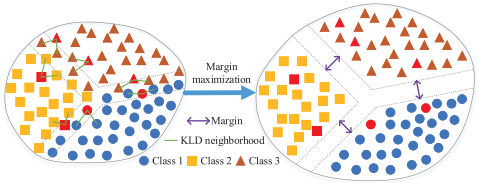



Fig. 5. Construction of weighted neighborhood margin.

interclass samples. The negative weights denote the interclass samples that their relationships should be suppressed in low-dimensional space. $d_{ij} = ||\mathbf{x}_i - \mathbf{x}_j||$ represents the distance between $\mathbf{x}_i$ and $\mathbf{x}_j$. For the supervised locality graph, we should gather the positive similarity and separate the negative similarity among all the vertices in low-dimensional embedding space. The optimization function can be defined as

$$\min_{\mathbf{V}} \frac{1}{2} \sum_{i,j=1}^{n} w_{ij}^{s} \left\| \frac{\mathbf{V}^T \mathbf{x}_i}{\sqrt{f_i}} - \frac{\mathbf{V}^T \mathbf{x}_j}{\sqrt{f_j}} \right\|^2 \tag{25}$$

where $f_i = \sum_{j=1}^{n} w_{i,j}^{s}$ is the degree of vertex $\mathbf{x}_i$ in the supervised locality graph. Based on some mathematical operations, (25) can be simplified as the following:

$$\begin{aligned}
&\frac{1}{2} \sum_{i,j=1}^{n} w_{ij}^{s} \left\| \frac{\mathbf{V}^T \mathbf{x}_i}{\sqrt{f_i}} - \frac{\mathbf{V}^T \mathbf{x}_j}{\sqrt{f_j}} \right\|^2 \\
&= \sum_{i,j=1}^{n} \operatorname{tr} \left( \mathbf{V}^T \mathbf{x}_i \frac{w_{ij}^{s}}{f_i} \mathbf{x}_i^T \mathbf{V} - \mathbf{V}^T \mathbf{x}_i \frac{w_{ij}^{s}}{\sqrt{f_i f_j}} \mathbf{x}_j^T \mathbf{V} \right) \\
&= \operatorname{tr} \left( \mathbf{V}^T \left\{ \sum_{i=1}^{n} \left[ \mathbf{x}_i \left( \frac{\sum_{j=1}^{n} w_{ij}^{s}}{f_i} \right) \mathbf{x}_i^T \right] \right. \right. \\
&\qquad \left. \left. - \sum_{i,j=1}^{n} \left( \mathbf{x}_i \frac{w_{ij}^{s}}{\sqrt{f_i f_j}} \mathbf{x}_j^T \right) \right\} \mathbf{V} \right) \\
&= \operatorname{tr} \left( \mathbf{V}^T \left[ \sum_{i=1}^{n} \left( \mathbf{x}_i \mathbf{x}_i^T \right) - \sum_{i,j=1}^{n} \left( \mathbf{x}_i f_i^{-1/2} w_{ij}^{s} f_j^{-1/2} \mathbf{x}_j^T \right) \right] \mathbf{V} \right) \\
&= \operatorname{tr}(\mathbf{V}^T \mathbf{X} (\mathbf{I} - \mathbf{F}^{-1/2} \mathbf{W}^s \mathbf{F}^{-1/2}) \mathbf{X}^T \mathbf{V}) \\
&= \operatorname{tr}(\mathbf{V}^T \mathbf{X} \mathbf{L}^s \mathbf{X}^T \mathbf{V})
\end{aligned} \tag{26}$$

where $\mathbf{L}^s = \mathbf{I} - \mathbf{F}^{-1/2} \mathbf{W}^s \mathbf{F}^{-1/2}$ denotes the supervised locality Laplacian matrix, $\mathbf{W}^s = [w_{ij}^{s}]_{i,j}$ is the weight matrix, and $\mathbf{F} = \operatorname{diag}(f_1, f_2, \ldots, f_n)$ is the diagonal matrix of the vertex degree. Therefore, with (26), the optimization problem for the supervised locality graph can be denoted as

$$\min_{\mathbf{V}} \operatorname{tr}(\mathbf{V}^T \mathbf{X} \mathbf{L}^s \mathbf{X}^T \mathbf{V}) \tag{27}$$

### C. Weighted Neighborhood Margin

To further enhance the difference of the interclass samples, we consider the neighborhood information to construct a weighted neighborhood margin based on the idea of the maximum margin. First, the KLD is adopted to obtain the neighbors of each sample, which is used to measure the difference of the probability distribution. Then, we use the weighted neighborhood of each sample to compute the interclass distances. Finally, in low-dimensional feature space, we maximize the summation of the marginal distances to enhance the discriminant performance of the interclass samples. The idea of weighted neighborhood margin is shown in Fig. 5 In this article, we use the KLD to reveal some other information that are different the neighborhood information of the two hypergraph models. The KLD is asymmetry, that is to say, $KL(\mathbf{x}_i||\mathbf{x}_j)$ is not equal to $KL(\mathbf{x}_j||\mathbf{x}_i)$, where $KL(\mathbf{x}_i||\mathbf{x}_j)$ is the KLD between $\mathbf{x}_i$ and $\mathbf{x}_j$. To maintain the symmetry, a symmetric KLD is denoted as the following:

$$D_{ij}^{p} = \frac{1}{2}(KL(\mathbf{x}_i||\mathbf{x}_j) + KL(\mathbf{x}_j||\mathbf{x}_i)) \tag{28}$$

where $D_{ij}^{p}$ is the symmetric KLD between $\mathbf{x}_i$ and $\mathbf{x}_j$, and $KL(\mathbf{p}||\mathbf{q})$ is denoted as

$$KL(\mathbf{p}||\mathbf{q}) = \sum\nolimits_i \left[ \left( p_i / \sum\nolimits_i p_i \right) \log \left( \frac{p_i / \sum_i p_i}{q_i / \sum_i q_i} \right) \right] \tag{29}$$

where $\mathbf{p} = [p_1, p_2, \ldots, p_D]$ and $\mathbf{q} = [q_1, q_2, \ldots, q_D]$ are two vectors. According to the symmetric KLD, we can obtain the $k_w$ intraclass neighbors of $x_i$ and its $k_b$ interclass neighbors. Therefore, with the KLD, the intraclass and interclass weighted values are represented as

$$w_{ij}^{pw} = \begin{cases} \exp\left( -\frac{\left(D_{ij}^{p}\right)^2}{2\left(t_i^{pw}\right)^2} \right), & \text{if } \mathbf{x}_j \in \mathcal{N}_{k_w}^{p}(\mathbf{x}_i) \text{ and } \ell(\mathbf{x}_i) = \ell(\mathbf{x}_j) \\ 0, & \text{otherwise} \end{cases} \tag{30}$$

$$w_{ik}^{pb} = \begin{cases} \exp\left( -\frac{\left(D_{ik}^{p}\right)^2}{2\left(t_i^{pb}\right)^2} \right), & \text{if } \mathbf{x}_k \in \mathcal{N}_{k_b}^{p}(\mathbf{x}_i) \text{and } \ell(\mathbf{x}_i) \neq \ell(\mathbf{x}_j) \\ 0, & \text{otherwise} \end{cases} \tag{31}$$

where $\mathcal{N}_{k_w}^{p}(\mathbf{x}_i)$ and $\mathcal{N}_{k_b}^{p}(\mathbf{x}_i)$ are the intraclass and interclass KLD NNs, respectively. $t_i^{pw} = (1/k_w) \sum_{\mathbf{x}_j \in \mathcal{N}_{k_w}^{p}(\mathbf{x}_i)} D_{ij}^{p}$ and $t_i^{pb} = (1/k_b) \sum_{\mathbf{x}_k \in \mathcal{N}_{k_b}^{p}(\mathbf{x}_i)} D_{ik}^{p}$ are two kernel parameters.

Then, in low-dimensional space, we should separate the weighted neighborhood margin as far as possible and the objective function can be defined as the following:

$$\max_{\mathbf{V}} \sum_{i=1}^{n} \left( \sum_{j=1}^{n} \frac{||\mathbf{V}^T \mathbf{x}_i - \mathbf{V}^T \mathbf{x}_j||^2 w_{ij}^{pb}}{\sum_{j=1}^{n} w_{ij}^{pb}} - \sum_{k=1}^{n} \frac{||\mathbf{V}^T \mathbf{x}_i - \mathbf{V}^T \mathbf{x}_k||^2 w_{ik}^{pw}}{\sum_{k=1}^{n} w_{ik}^{pw}} \right). \tag{32}$$

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With some mathematical operations, (32) can be simplified using the following process:

$$\begin{aligned}\sum_{i=1}^{n}\sum_{j=1}^{n}\frac{||\mathbf{V}^T\mathbf{x}_i-\mathbf{V}^T\mathbf{x}_j||^2 w_{ij}^{pb}}{\sum_{j=1}^{n} w_{ij}^{pb}} &= \mathrm{tr}\left(\sum_{i=1}^{n}\sum_{j=1}^{n}\frac{(\mathbf{V}^T\mathbf{x}_i-\mathbf{V}^T\mathbf{x}_j)w_{ij}^{pb}(\mathbf{V}^T\mathbf{x}_i-\mathbf{V}^T\mathbf{x}_j)^T}{\sum_{j=1}^{n} w_{ij}^{pb}}\right)\\ &= \mathrm{tr}\left(\mathbf{V}^T\sum_{i=1}^{n}\sum_{j=1}^{n}\frac{(\mathbf{x}_i-\mathbf{x}_j)w_{ij}^{pb}(\mathbf{x}_i-\mathbf{x}_j)^T}{\sum_{j=1}^{n} w_{ij}^{pb}}\mathbf{V}\right)\\ &= \mathrm{tr}(\mathbf{V}^T\mathbf{S}_b\mathbf{V}) \qquad (33)\end{aligned}$$

$$\sum_{i=1}^{n}\sum_{k=1}^{n}\frac{||\mathbf{V}^T\mathbf{x}_i-\mathbf{V}^T\mathbf{x}_k||^2 w_{ik}^{pw}}{\sum_{k=1}^{n} w_{ik}^{pw}} = \mathrm{tr}(\mathbf{V}^T\mathbf{S}_w\mathbf{V}) \qquad (34)$$

where $\mathbf{S}_b$ and $\mathbf{S}_w$ are the interclass and intraclass weighted neighborhood scatter matrices, and they are defined as

$$\mathbf{S}_b = \sum_{i=1}^{n}\sum_{j=1}^{n}\frac{(\mathbf{x}_i-\mathbf{x}_j)w_{ij}^{pb}(\mathbf{x}_i-\mathbf{x}_j)^T}{\sum_{j=1}^{n} w_{ij}^{pb}} \qquad (35)$$

$$\mathbf{S}_w = \sum_{i=1}^{n}\sum_{k=1}^{n}\frac{(\mathbf{x}_i-\mathbf{x}_k)w_{ik}^{pw}(\mathbf{x}_i-\mathbf{x}_k)^T}{\sum_{k=1}^{n} w_{ik}^{pw}}. \qquad (36)$$

Therefore, in low-dimensional embedding space, the weighted neighborhood margin function can be rewritten as follows:

$$\max_{\mathbf{V}}\ \mathrm{tr}(\mathbf{V}^T(\mathbf{S}_b-\mathbf{S}_w)\mathbf{V}) = \max_{\mathbf{V}}\ \mathrm{tr}(\mathbf{V}^T\mathbf{S}\mathbf{V}) \qquad (37)$$

where $\mathbf{S}=\mathbf{S}_b-\mathbf{S}_w$ is the weighted neighborhood margin scatter matrix.

### D. Feature Learning and Optimization

In this section, we propose an EHGDL method to gather the merits of graph and hypergraph, and we can obtain the objective function of EHGDL as the following:

$$\max_{\mathbf{V}}\frac{\mathrm{tr}[\mathbf{V}^T(\mathbf{X}\mathbf{L}^b\mathbf{X}^T+\alpha\mathbf{S})\mathbf{V}]}{\mathrm{tr}[\mathbf{V}^T(\mathbf{X}\mathbf{L}^w\mathbf{X}^T+\beta\mathbf{X}\mathbf{L}^s\mathbf{X}^T)\mathbf{V}]} \qquad (38)$$

where $\alpha>0$ and $\beta>0$ are the trade-off parameters. For the optimization (38), we can transform it to the following optimal problem according to the principle of generalized Rayleigh quotient [68]:

$$\begin{aligned}&\max_{\mathbf{V}}\mathbf{V}^T(\mathbf{X}\mathbf{L}^b\mathbf{X}^T+\alpha\mathbf{S})\mathbf{V}\\ &\text{s.t. }\mathbf{V}^T(\mathbf{X}\mathbf{L}^w\mathbf{X}^T+\beta\mathbf{X}\mathbf{L}^s\mathbf{X}^T)\mathbf{V}=\mathbf{I}. \qquad (39)\end{aligned}$$

According to the method of the Lagrangian multipliers, we can obtain the Lagrangian function as the following with the Lagrangian multiplier $\lambda$:

$$\begin{aligned}L(\mathbf{V},\lambda) &= \mathbf{V}^T(\mathbf{X}\mathbf{L}^b\mathbf{X}^T+\alpha\mathbf{S})\mathbf{V}\\ &\quad-\lambda[\mathbf{V}^T(\mathbf{X}\mathbf{L}^w\mathbf{X}^T+\beta\mathbf{X}\mathbf{L}^s\mathbf{X}^T)\mathbf{V}-\mathbf{I}]. \qquad (40)\end{aligned}$$

To achieve the optimal solution, we calculate the derivative of $\mathbf{V}$, that is

$$\frac{\partial L(\mathbf{V},\lambda)}{\partial\mathbf{V}} = (\mathbf{X}\mathbf{L}^b\mathbf{X}^T+\alpha\mathbf{S})\mathbf{V}-\lambda(\mathbf{X}\mathbf{L}^w\mathbf{X}^T+\beta\mathbf{X}\mathbf{L}^s\mathbf{X}^T)\mathbf{V}. \qquad (41)$$

Then, let $\partial L(\mathbf{V},\lambda)/\partial\mathbf{V}=0$, we can obtain a generalized eigenvalue problem for the optimization of (38) as follows:

$$(\mathbf{X}\mathbf{L}^b\mathbf{X}^T+\alpha\mathbf{S})\mathbf{v}_i = \lambda_i(\mathbf{X}\mathbf{L}^w\mathbf{X}^T+\beta\mathbf{X}\mathbf{L}^s\mathbf{X}^T)\mathbf{v}_i \qquad (42)$$

where $\lambda_i$ is an eigenvalue and its corresponding eigenvector is $\mathbf{v}_i$. For the generalized eigenvalue problems (42), we can achieve the projection matrix $\mathbf{V}$ with the eigenvectors of $d$ largest eigenvalues

$$\mathbf{V}=[\mathbf{v}_1,\mathbf{v}_2,\ldots,\mathbf{v}_d]\in\Re^{D\times d}. \qquad (43)$$

According to the projection matrix, we can get the low-dimensional embedding features of $\mathbf{x}_i$ as

$$\mathbf{y}_i=\mathbf{V}^T\mathbf{x}_i. \qquad (44)$$

### E. Illustration of EHGDL

In general, the proposed EHGDL method constructs an intraclass hypergraph and a supervised locality graph to achieve the compactness of the intraclass samples in low-dimensional space, while it also designs an interclass hypergraph and a weighted neighborhood margin to enhance the separability of the interclass samples in low-dimensional space. EHGDL not only reveals the complex high-order properties but also represents the binary characteristics, which can achieve the complementarity of different structures of data and enhance the discriminant performance of features for classification. The steps of the proposed method is shown in Algorithm 1.

According to Algorithm 1, we can analyze the computational complexity of EHGDL with a big $O$ notation. To construct the intraclass hypergraph, it takes $O(n^2)$ to achieve $\mathbf{H}^w$, $\mathbf{D}^{vw}$, $\mathbf{D}^{ew}$, and $O(nk_w)$ to obtain $\mathbf{W}^w$. The cost of $\mathbf{L}^w$ is $O(n^3)$. Thus, the total cost is $O(n^3+3n^2+nk_w)$ for the representation of the intraclass high-order information. To reveal the interclass high-order properties from the interclass hypergraph, the computational complexity is $O(n^3+3n^2+nk_b)$ with the same analysis as the intraclass hypergraph. For the supervised locality graph, the coding coefficients and the weight matrix $\mathbf{W}^s$ both require the cost of $O(n^2)$. For the supervised locality Laplacian matrix $\mathbf{L}^s$, the cost is $O(n^2)$. So the total cost is $O(3n^2)$ for the supervised locality information. For the weighted neighborhood margin, it both takes $O(2n^2)$ to compute the intraclass and interclass weighted neighborhood scatter matrices. The weighted neighborhood margin scatter matrix $\mathbf{S}$ has the complexity of $O(n^2)$. Then, we can obtain the total cost of $O(5n^2)$ to process the margin information. To compute the matrices $\mathbf{J}_1=\mathbf{X}\mathbf{L}^b\mathbf{X}^T+\alpha\mathbf{S}$ and $\mathbf{J}_2=\mathbf{X}\mathbf{L}^w\mathbf{X}^T+\beta\mathbf{X}\mathbf{L}^s\mathbf{X}^T$, they need the cost of $O(Dn^2+D^2)$ and $O(2Dn^2+D^2)$, respectively. In addition, the cost is $O(D^3)$ to solve the generalized eigenvalue. Therefore, we can obtain that the total computational complexity of EHGDL is $O(n^3+Dn^2+D^3)$, which mainly depends on the size of the input samples and their dimensionality.

## IV. Experimental Results and Discussion

To show the effectiveness of EHGDL, we designed some experiments on three popular HSI data sets (Indian Pines, PaviaU, and HoustonU) and selected several state-of-the-art methods for comparison.

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**Algorithm 1** EHGDL

**Input:** Samples $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2, \cdots, \mathbf{x}_n] \in \Re^{D\times n}$ and the corresponding class label $\{\ell(\mathbf{x}_1), \ell(\mathbf{x}_2), \cdots, \ell(\mathbf{x}_n)\}$, intraclass weighted parameter $\theta$, trade-off parameters $\alpha$ and $\beta$, interclass neighbor size $k_b$, intraclass neighbor size $k_w$, and embedding dimension $d$.

**Output:** Low-dimensional embedding features $\mathbf{Y} = [\mathbf{y}_1, \mathbf{y}_2, \cdots, \mathbf{y}_n] \in \Re^{d\times n}$.

Construct the intraclass hypergraph with the $k_w$ intraclass neighbors using Euclidean distance.

Compute the intraclass hyper-Laplacian matrix:

$$\mathbf{L}^w = \mathbf{I} - (\mathbf{D}^{vw})^{-1/2}\mathbf{H}^w\mathbf{W}^w(\mathbf{D}^{ew})^{-1}(\mathbf{H}^w)^T(\mathbf{D}^{vw})^{-1/2}$$

Construct the interclass hypergraph with the $k_b$ interclass neighbors using Euclidean distance.

Compute the interclass hyper-Laplacian matrix:

$$\mathbf{L}^b = \mathbf{I} - (\mathbf{D}^{vb})^{-1/2}\mathbf{H}^b\mathbf{W}^b(\mathbf{D}^{eb})^{-1}(\mathbf{H}^b)^T(\mathbf{D}^{vb})^{-1/2}$$

Construct the supervised locality graph with the locality-constrained coding coefficients and the label information.

Compute the supervised locality Laplacian matrix:

$$\mathbf{L}^s = \mathbf{I} - \mathbf{F}^{-1/2}\mathbf{W}^s\mathbf{F}^{-1/2}$$

Select the $k_w$ intraclass neighbors to construct margin according to the Kullback-Leibler divergence.

Compute the intraclass weighted neighborhood scatter matrix:

$$\mathbf{S}_w = \sum_{i=1}^{n}\sum_{k=1}^{n}\frac{(\mathbf{x}_i-\mathbf{x}_k)w_{ik}^{pw}(\mathbf{x}_i-\mathbf{x}_k)^T}{\sum_{k=1}^{n}w_{ik}^{pw}}$$

Select the $k_b$ interclass neighbors to construct margin according to the Kullback-Leibler divergence.

Compute the interclass weighted neighborhood scatter matrix:

$$\mathbf{S}_b = \sum_{i=1}^{n}\sum_{j=1}^{n}\frac{(\mathbf{x}_i-\mathbf{x}_j)w_{ij}^{pb}(\mathbf{x}_i-\mathbf{x}_j)^T}{\sum_{j=1}^{n}w_{ij}^{pb}}$$

Obtain the weighted neighborhood margin scatter matrix:

$$\mathbf{S} = \mathbf{S}_b - \mathbf{S}_w$$

Solve the generalized eigenvalue problem:

$$\left(\mathbf{X}\mathbf{L}^b\mathbf{X}^T + \alpha\mathbf{S}\right)\mathbf{v}_i = \lambda_i\left(\mathbf{X}\mathbf{L}^w\mathbf{X}^T + \beta\mathbf{X}\mathbf{L}^s\mathbf{X}^T\right)\mathbf{v}_i$$

Obtain the projection matrix with $d$ largest eigenvalues corresponding eigenvectors:

$$\mathbf{V} = [\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_d] \in \Re^{D\times d}$$

Calculate the low-dimensional embedding features:

$$\mathbf{y}_i = \mathbf{V}^T\mathbf{x}_i$$

### A. Data Set Description

*1) Indian Pines Data Set [69]:* This data set is a scene of the Northwest Indiana collected by an Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor in 1992. These data are composed of 145 × 145 pixels and 220 spectral bands within the range of 375–2500 nm. Several spectral bands, including bands 104–108, 150–163, and 220, with noise and water absorption phenomena, were removed from the data set, leaving a total of 200 radiance channels to be used in the experiments. The scene in false color and its corresponding groundtruth are shown in Fig. 6.

*2) PaviaU Data Set [69]:* The HSI data set was acquired with a Reflective Optics System Imaging Spectrometer (ROSIS) optical sensor during a flight campaign over Pavia University (PaviaU), Northern Italy, in 2002. The data set contains 610 × 340 pixels, and the geometric resolution was 1.3 m. Exactly 103 spectral bands were remained after the removal of some channels because of the noise effect. For illustrative purposes, the HSI in false color and its corresponding groundtruth are shown in Fig. 7.

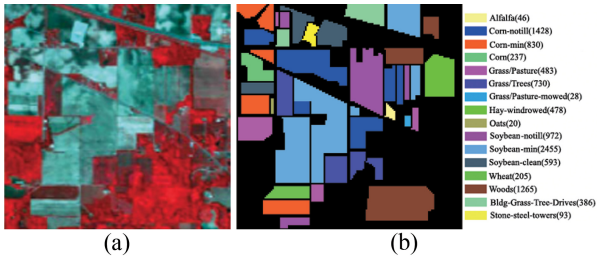



Fig. 6. Indian Pines HSI. (a) HSI in false color. (b) Groundtruth. (Note that the number of samples for each class is shown in brackets.)

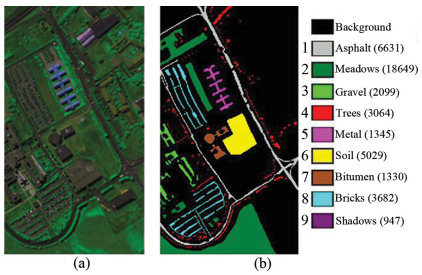



Fig. 7. PaviaU HSI. (a) HSI in false color. (b) Corresponding groundtruth. (Note that the number of samples for each class is shown in brackets.)

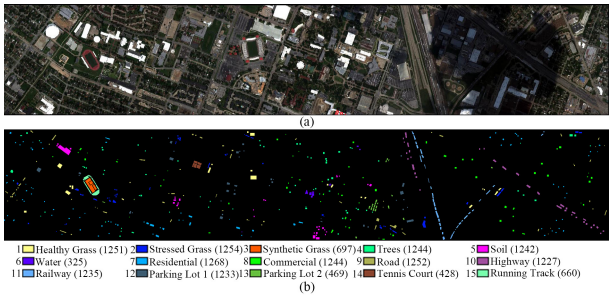



Fig. 8. PaviaU HSI. (a) HSI in false color. (b) Corresponding groundtruth. (Note that the number of samples for each class is shown in brackets.)

*3) HoustonU Data Set [70]:* The data set was captured by an ITRES-CASI (Compact Airborne Spectrographic Imager) 1500 hyperspectral imager covering the University of Houston campus and its neighboring urban area. The spatial resolution of this data is 2.5 m and its size is 349 × 1905 pixels. These data contain 144 spectral bands spanning the visible and near-infrared spectrum from 380 to 1050 nm. It has 15 land-cover types and Fig. 8 displays the HSI in false color and its corresponding groundtruth.

### B. Experimental Setup

In the experiments, we randomly selected some samples as training set and the remaining samples as test set. For the

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classes with small samples (i.e., Alfalfa, Grass/Pasture-mowed, and Oats from the Indian Pines data set), we randomly selected 10 samples as training set. First, we use the training set to train each DR model. Then, the low-dimensional embedding features of all the samples can be obtained with these training models. Finally, we select a classifier to discriminate the class type of each test sample, meanwhile applying the classification accuracy of each class, the average accuracy of all classes (AA), the overall classification accuracy (OA), and the kappa coefficient (KC) to evaluate the classification results of different methods.

In this article, we proposed a DR method, thus we selected some relevant DR algorithms to compare with EHGDL. To analyze the hypergraph model, we selected two hypergraph DR methods, termed BH [58] and DHLP [57], for comparison. To evaluate the margin model, the MMC [34] method is selected as a compared method. To demonstrate the discriminating power, we compare EHGDL with three discriminant methods including LFDA [41], RLDE [42], and LGSFA [43]. In addition, we also apply some advanced graph learning methods, i.e. SDE [52], SGDA [48], CGDA [54], LGDA [53], and SLGDA [53], to show the effectiveness of EHGDL. After obtaining the low-dimensional embedding features, we selected three simple classifiers, which do not need to adjust any parameters, to obtain the class labels of the test set, such as the NN classifier, the spectral angle mapper (SAM) and the maximum likelihood classifier (MLC). Simultaneously, we also displayed the classification results of the raw spectral feature (RAW) with NN, SAM, and MLC.

In the experiments, we randomly selected 10%, 5%, and 10% samples from each class on the Indian Pines, PaviaU, and HoustonU data sets as training samples, and the other samples were applied for the test. For the compared algorithms, the grid search method was used to obtain the optimal parameters. For the proposed method, we set the trade-off parameters $\alpha$ and $\beta$ both from 0 to 4 at an interval of 0.2 for the search of the optimization value. The optimized intraclass and interclass neighbors $k_w$ and $k_b$ of EHGDL were selected at a range of 3–31 at an interval of 2. For the intraclass weighted parameter $\theta$, we obtained the optimization results from 1 to 20 at an interval of 1. Therefore, we set the $\alpha$, $\beta$, $k_w$, $k_b$ and $\theta$ to 0.8, 1.6, 13, 13, and 15 for the Indian Pines data set, 3.2, 2.2, 19, 21, and 1 for the PaviaU data set, and 3, 1, 15, 23, and 9 for the HoustonU data set. For all the methods, the embedding dimension is set to 25. To robustly analyze each algorithm, we repeated each experiment 10 times in each condition and obtained the average OAs with standard deviations (STDs) and the average KCs. All the experiments were conducted on a personal computer with E5-2620 v3 central processing unit, 16-GB memory, and 64-bit Windows 10 using MATLAB 2017b, and we gave the running time including DR time (T1) and classification time (T2).

### *C. Comparison With Different Discriminant Models*

To demonstrate the effectiveness of different discriminant models, we conducted some experiments on the three data sets with different discriminant models.

According to the intraclass and interclass hypergraphs, we construct a hypergraph discriminant learning (HDL) method to compact the similarity of the intraclass hypergraph and separate the characteristics of the interclass hypergraph in low-dimensional embedding space. The objective function of HDL is given as the following:

$$\max_{\mathbf{V}} \frac{\mathrm{tr}(\mathbf{V}^T\mathbf{X}\mathbf{L}^b\mathbf{X}^T\mathbf{V})}{\mathrm{tr}(\mathbf{V}^T\mathbf{X}\mathbf{L}^w\mathbf{X}^T\mathbf{V})}. \tag{45}$$

To enhance the discriminant performance of HDL, we design an enhanced HDL (EHDL) method with the weighted neighborhood margin to increase the difference of the low-dimensional embedding samples from different classes, and an optimization function of EHDL is defined as follows:

$$\max_{\mathbf{V}} \frac{\mathrm{tr}[\mathbf{V}^T(\mathbf{X}\mathbf{L}^b\mathbf{X}^T+\alpha\mathbf{S})\mathbf{V}]}{\mathrm{tr}(\mathbf{V}^T\mathbf{X}\mathbf{L}^w\mathbf{X}^T\mathbf{V})} \tag{46}$$

where $\alpha > 0$ is the trade-off parameters.

To effectively reveal the complex intrinsic structures of HSI, we construct a complementary model of graph and hypergraph, termed hybrid-graph discriminant learning (HGDL), on the basis of the interclass hypergraph, the interclass hypergraph, and the supervised locality graph. The feature embedding model is represented as the following:

$$\max_{\mathbf{V}} \frac{\mathrm{tr}(\mathbf{V}^T\mathbf{X}\mathbf{L}^b\mathbf{X}^T\mathbf{V})}{\mathrm{tr}[\mathbf{V}^T(\mathbf{X}\mathbf{L}^w\mathbf{X}^T+\beta\mathbf{X}\mathbf{L}^s\mathbf{X}^T)\mathbf{V}]} \tag{47}$$

where $\beta > 0$ is the trade-off parameters.

Then, with the interclass hypergraph and the supervised locality graph, we construct a between-class HGDL (BHGDL) method to compact the intraclass information of the graph and separate the interclass information of the hypergraph in low-dimensional embedding space. The objective function of BHGDL is defined as

$$\max_{\mathbf{V}} \frac{\mathrm{tr}(\mathbf{V}^T\mathbf{X}\mathbf{L}^b\mathbf{X}^T\mathbf{V})}{\mathrm{tr}(\mathbf{V}^T\mathbf{X}\mathbf{L}^s\mathbf{X}^T\mathbf{V})}. \tag{48}$$

To enhance the performance of BHGDL, we develop an enhanced BHGDL (EBHGDL) method with the weighted neighborhood margin to improve the discriminant performance of different classes, and the embedding function is defined as follows:

$$\max_{\mathbf{V}} \frac{\mathrm{tr}[\mathbf{V}^T(\mathbf{X}\mathbf{L}^b\mathbf{X}^T+\alpha\mathbf{S})\mathbf{V}]}{\mathrm{tr}(\mathbf{V}^T\mathbf{X}\mathbf{L}^s\mathbf{X}^T\mathbf{V})} \tag{49}$$

where $\alpha > 0$ is the trade-off parameters. According to the intraclass hypergraph, the supervised locality graph, and the weighted neighborhood margin, we develop an enhanced within-class HGDL (EWHGDL) method to improve the compactness of different intraclass information, and the objective function is represented as

$$\max_{\mathbf{V}} \frac{\mathrm{tr}(\mathbf{V}^T\mathbf{S}\mathbf{V})}{\mathrm{tr}[\mathbf{V}^T(\mathbf{X}\mathbf{L}^w\mathbf{X}^T+\beta\mathbf{X}\mathbf{L}^s\mathbf{X}^T)\mathbf{V}]} \tag{50}$$

where $\beta > 0$ is the trade-off parameters. In addition, we consider the weighted neighborhood margin and the supervised

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TABLE I
CLASSIFICATION RESULTS UNDER DIFFERENT DISCRIMINANT MODELS ON THE THREE DATA SETS (OA±STD (KC)%)

| Method | Indian Pines | PaviaU | HoustonU |
|---|---|---|---|
| HDL | 74.38±0.69 (70.71) | 87.69±0.72 (83.57) | 94.67±0.37 (94.23) |
| BHGDL | 72.84±1.80 (68.95) | 89.98±0.59 (86.59) | 94.16±0.47 (93.69) |
| HGDL | 78.68±1.19 (75.64) | 89.12±0.56 (85.49) | 94.97±0.22 (94.56) |
| EGDL | 73.48±4.37 (69.68) | 90.56±0.65 (87.41) | 94.15±0.61 (93.68) |
| EWHDL | 74.57±1.08 (70.84) | 77.42±1.61 (69.95) | 93.97±0.62 (93.48) |
| EHDL | 75.75±0.90 (72.23) | 88.61±0.58 (84.69) | 95.35±0.39 (94.97) |
| EWHGDL | 80.65±0.61 (77.87) | 89.39±0.58 (85.86) | 94.83±0.48 (94.41) |
| EBHGDL | 73.93±1.34 (70.22) | 89.90±0.39 (86.46) | 94.20±0.34 (93.72) |
| EHGDL | **81.01**±0.68 (**78.28**) | **90.77**±0.39 (**87.65**) | **95.57**±0.59 (**95.20**) |

locality graph structure to construct an enhanced graph discriminant learning (EGDL) method. The embedding function can be represented as

$$\max_{\mathbf{V}} \frac{\mathrm{tr}(\mathbf{V}^T\mathbf{S}\mathbf{V})}{\mathrm{tr}(\mathbf{V}^T\mathbf{X}\mathbf{L}^s\mathbf{X}^T\mathbf{V})}. \tag{51}$$

Finally, we use the weighted neighborhood margin and the intraclass hypergraph to develop an enhanced within-class HDL (EWHDL) method and the objective function is

$$\max_{\mathbf{V}} \frac{\mathrm{tr}(\mathbf{V}^T\mathbf{S}\mathbf{V})}{\mathrm{tr}(\mathbf{V}^T\mathbf{X}\mathbf{L}^w\mathbf{X}^T\mathbf{V})}. \tag{52}$$

In the experiments, for EHDL and EBHGDL, the parameter $\alpha$ is set to 0.8 for the Indian Pines data set, 3.2 for the PaviaU data set, and 3 for the HoustonU data set. For HGDL, the parameter $\beta$ is set to 0.1 for the HoustonU and PaviaU data sets, and 0.6 for the Indian Pines data set. For EWHGDL, the parameter $\beta$ is set to 1.6 for the Indian Pines data set, 2.2 for the PaviaU data set, and 1 for the HoustonU data set. Table I shows the OAs with STD and the KCs of different discriminant models With NN.

In Table I, HGDL achieves higher classification accuracies compared with HDL on the three data sets, because HGDL uses not only the hypergraph properties but also the graph characteristics which can form the complementarity of different intrinsic structures to represent the embedding features. To analyze EWHDL and EHDL, EHDL generates better classification than EWHDL on the three data sets, since EHDL considers the interclass hypergaph model on the basis of EWHDL to enhance the separability of samples from different classes. EHDL is superior to EGDL on the Indian Pines and HoustonU data sets for the hypergraph can reveal high-order information to enhance the homogeneity of the related samples. While EGDL has better classification performance than EHDL on the PaviaU data set, as the PaviaU data set possesses more similar spectrum in different classes than the other two data sets which may cause that the samples from different classes are difficult to discriminate in the hypergraph. EHGDL and EHDL, respectively, obtain better classification results than HGDL and HDL on the three data sets, which indicate that the weighted neighborhood margin can enhance the discriminant performance of different classes. EHGDL inherits the merits of margin, hypergraph, and graph to better reveal the intrinsic properties of HSI and effectively extract the discriminant features, thus it obtains the best classification results compared with the other discriminant models. As a result, the weighted neighborhood margin can enhance the discriminant performance of different classes, and the supervised locality graph can improve the heterogeneity of the same class samples. Furthermore, the proposed EHGDL can promote the compactness of the intraclass samples and the separability of the interclass samples in low-dimensional embedding space.

### D. Classification Results and Running Time

*1) Indian Pines:* To compare the classification with the other state-of-the-art method, we applied the training set with 10% samples from each class to learn a projection matrix of each method and extracted the low-dimensional embedding features of all the samples. After that, we adopted the NN classifier to achieve the classes of the test samples. The results and running time are shown in Table II, and Fig. 9 displays the corresponding classification maps.

*2) PaviaU:* To further evaluate the effectiveness of EHGDL, we selected another scene to perform the compared experiments with 5% samples from each class, and the classification results with NN are given in Table III. Fig. 10 is the visualized classification results of Table III.

*3) HoustonU:* To demonstrate the performance of the proposed method on a large scale scene, we applied the HoustonU data set to do some experiments with 10% samples from each class. Table IV shows the each class accuracies, AAs, OAs, KCs and running time of different methods with the NN classifier, and the classification maps of different methods are given in Fig. 11.

According to Tables II–IV, the proposed method generates better classification accuracies in most classes compared with the other related methods on the Indian Pines, PaviaU, and HoustonU data sets. EHGDL achieves better AAs, OAs, and KCs than the other hypergraph learning algorithms (i.e., BH and DHLP) on the three data sets, because EHGDL applies the marginal information between different classes and the locality-constrained relationship between two samples to enhance the discriminant performance of the low-dimensional embedding features. EHGDL improves the performance of the land-cover classification on the three data sets to compare with the graph learning methods, including SDE, SGDA, CGDA, LGDA, and SLGDA, since the proposed method considers the complex multiple relationships of HSI that is benefit to reveal the similar properties of data and enhance the difference of the low-dimensional embedding features between different classes. We also compare EHGDL with a margin algorithm (MMC) on the three data sets and EHGDL obtains better classification performance for the reason that our method analyzes not only the marginal information but also the binary and multiple relationships of data. In addition, EHGDL achieves better OAs, AAs, and KCs on the three data sets than some

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TABLE II

CLASSIFICATION ACCURACIES OF EACH CLASS (%) AND RUNNING TIME (SECOND) FOR DIFFERENT METHODS ON THE INDIAN PINES DATA SET

| Class | RAW | MMC [34] | LFDA [41] | LGSFA [43] | RLDE [42] | BH [57] | DHLP [56] | SDE [51] | SGDA [47] | CGDA [53] | LGDA [52] | SLGDA [52] | EHGDL |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 76.09 | 80.43 | 76.09 | 78.26 | 58.70 | 69.57 | 82.61 | 60.87 | 67.39 | 73.91 | 67.39 | 67.39 | **84.78** |
| 2 | 57.91 | 58.33 | 63.59 | 64.15 | **69.40** | 58.61 | 62.54 | 57.35 | 54.48 | 55.81 | 54.55 | 54.55 | 60.29 |
| 3 | 54.94 | 52.65 | 53.49 | 65.42 | 47.35 | 57.95 | 64.82 | 56.63 | 62.77 | 65.66 | 64.22 | 63.86 | **75.78** |
| 4 | 39.66 | 43.04 | 39.24 | 47.26 | 18.99 | 48.10 | 51.90 | 40.93 | 35.44 | 36.71 | 48.10 | 47.26 | **64.56** |
| 5 | 83.02 | 81.37 | 83.64 | 92.75 | 81.57 | 84.68 | **94.00** | 80.54 | 80.12 | 80.33 | 90.06 | 89.65 | 87.37 |
| 6 | 91.10 | 90.27 | 90.82 | 93.29 | 85.62 | 87.40 | **94.52** | 88.90 | 89.04 | 91.37 | 92.33 | 92.60 | 94.25 |
| 7 | 89.29 | 89.29 | 89.29 | 89.29 | 92.86 | 89.29 | 85.71 | 89.29 | 85.71 | **96.43** | 89.29 | 89.29 | 85.71 |
| 8 | 94.14 | 93.31 | 94.77 | 98.12 | **99.16** | 96.23 | 98.74 | 96.23 | 97.91 | 98.54 | 95.40 | 95.40 | 95.82 |
| 9 | 65.00 | 70.00 | 65.00 | 60.00 | **75.00** | 60.00 | 60.00 | 65.00 | 60.00 | 65.00 | 65.00 | 65.00 | 70.00 |
| 10 | 67.90 | 67.39 | 64.51 | 67.70 | 8.74 | 58.02 | 65.64 | 65.33 | 55.35 | 59.67 | 54.53 | 54.32 | **69.75** |
| 11 | 76.25 | 75.85 | 75.72 | 82.53 | 74.42 | 77.15 | 80.24 | 79.47 | 78.09 | 78.82 | 82.20 | 82.53 | **87.98** |
| 12 | 46.54 | 45.87 | 46.54 | 68.30 | 11.13 | 41.65 | 69.14 | 46.88 | 50.59 | 51.77 | 63.24 | 63.07 | **72.85** |
| 13 | 94.15 | 93.17 | 96.59 | 98.54 | 96.59 | 97.07 | **99.02** | 97.07 | 96.10 | 98.54 | 97.07 | 97.07 | 98.54 |
| 14 | 90.83 | 91.38 | 93.12 | 96.21 | **98.02** | 91.38 | 95.73 | 92.25 | 94.86 | 93.52 | 90.43 | 90.75 | 95.97 |
| 15 | 39.90 | 36.53 | 40.16 | 52.07 | 46.37 | 35.23 | **53.11** | 37.82 | 45.08 | 42.75 | 40.67 | 40.93 | 50.52 |
| 16 | 86.02 | 84.95 | 87.10 | 84.95 | 83.87 | 82.80 | 84.95 | **88.17** | 79.57 | 81.72 | 83.87 | 83.87 | 86.02 |
| AA | 72.05 | 72.11 | 72.48 | 77.43 | 65.49 | 70.95 | 77.67 | 71.42 | 70.78 | 73.16 | 73.65 | 73.60 | **80.01** |
| OA | 71.71 | 71.25 | 72.32 | 78.35 | 65.01 | 71.04 | 77.67 | 72.28 | 71.76 | 72.95 | 73.77 | 73.82 | **80.49** |
| KC | 67.65 | 67.10 | 68.30 | 75.16 | 59.21 | 66.87 | 74.40 | 68.21 | 67.60 | 68.98 | 69.88 | 69.93 | **77.58** |
| T1(s) | - | 0.08 | 0.09 | 5.91 | 3.59 | 2.19 | 6.10 | 6.41 | 0.58 | 0.50 | 24.17 | 372.82 | 100.14 |
| T2(s) | 45.54 | 8.61 | 5.97 | 7.41 | 6.25 | 8.75 | 8.91 | 8.51 | 6.27 | 8.38 | 8.59 | 6.05 | 8.93 |

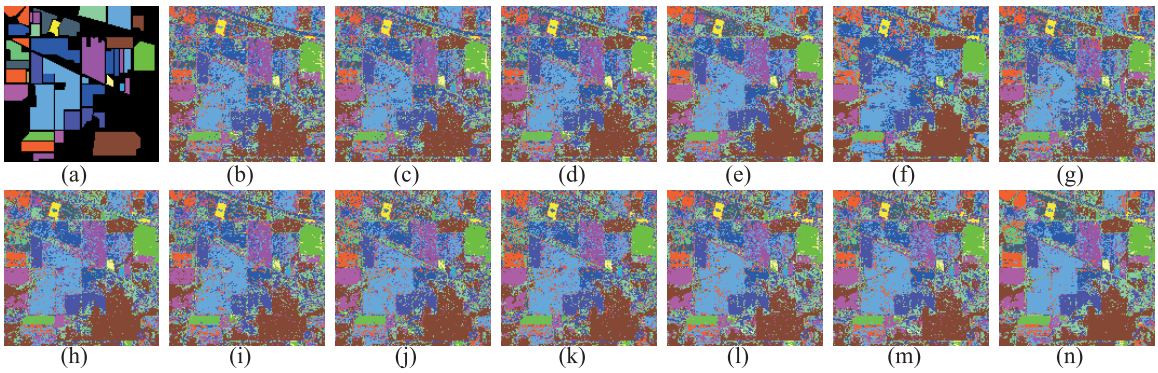



Fig. 9. Classification maps of different methods on the Indian Pines data set. (a) Groundtruth. (b) RAW. (c) MMC. (d) LFDA. (e) LGSFA. (f) RLDE. (g) BH. (h) DHLP. (i) SDE. (j) SGDA. (k) CGDA. (l) LGDA. (m) SLGDA. (n) EHGDL.

TABLE III

CLASSIFICATION ACCURACIES OF EACH CLASS (%) AND RUNNING TIME (SECOND) FOR DIFFERENT METHODS ON THE PAVIAU SET

| Class | RAW | MMC [34] | LFDA [41] | LGSFA [43] | RLDE [42] | BH [57] | DHLP [56] | SDE [51] | SGDA [47] | CGDA [53] | LGDA [52] | SLGDA [52] | EHGDL |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 86.28 | 86.26 | 87.06 | 87.33 | 89.87 | 89.59 | 85.12 | 87.95 | 72.28 | 78.98 | 88.07 | 87.23 | **92.76** |
| 2 | 94.88 | 94.76 | 96.28 | 98.16 | 97.74 | 95.87 | 94.92 | 96.09 | 94.85 | 94.66 | 96.29 | 96.26 | **98.21** |
| 3 | 65.75 | 65.46 | 69.99 | 63.84 | 70.03 | 71.37 | **73.46** | 70.51 | 66.60 | 65.41 | 69.03 | 70.65 | 72.13 |
| 4 | 78.33 | 78.39 | 80.12 | **88.61** | 85.15 | 80.16 | 85.57 | 78.23 | 86.29 | 71.93 | 87.40 | 86.65 | 87.96 |
| 5 | 98.88 | 98.88 | 99.18 | 99.85 | 99.55 | 99.33 | 99.85 | 99.18 | 99.85 | 99.85 | **99.93** | 99.85 | 99.85 |
| 6 | 62.24 | 62.26 | 66.91 | 62.42 | 72.00 | 68.12 | 73.16 | 67.21 | 68.03 | 70.63 | 76.26 | 76.73 | **77.85** |
| 7 | 76.62 | 76.69 | 79.92 | 72.26 | 81.58 | 82.48 | 78.57 | 83.01 | 63.38 | 67.82 | 77.52 | 73.83 | **83.38** |
| 8 | 81.23 | 81.07 | 81.80 | 77.95 | 83.19 | 82.81 | 71.97 | 80.93 | 50.65 | 63.31 | 77.24 | 76.94 | **84.03** |
| 9 | **100** | **100** | **100** | 98.10 | **100** | **100** | 97.99 | **100** | 51.21 | 66.31 | 91.87 | 89.02 | 99.79 |
| AA | 82.69 | 82.64 | 84.59 | 83.17 | 86.57 | 85.53 | 84.51 | 84.79 | 72.57 | 75.43 | 84.85 | 84.13 | **88.44** |
| OA | 85.59 | 85.52 | 87.37 | 87.42 | 89.59 | 87.97 | 86.86 | 87.37 | 80.61 | 82.35 | 88.48 | 88.21 | **91.36** |
| KC | 80.61 | 80.51 | 83.00 | 83.01 | 85.99 | 83.82 | 82.47 | 83.00 | 74.12 | 76.43 | 84.61 | 84.26 | **88.41** |
| T1(s) | - | 0.13 | 0.22 | 8.54 | 5.24 | 4.17 | 10.71 | 9.60 | 3.30 | 1.47 | 11.87 | 6085.41 | 320.34 |
| T2(s) | 342.12 | 84.46 | 69.97 | 83.79 | 83.70 | 83.79 | 84.20 | 84.10 | 85.38 | 106.66 | 84.96 | 69.57 | 116.67 |

discriminant algorithms including LFDA, RLDE, and LGSFA, which indicates that the proposed method can improve the discriminating power of data and enhance the compactness of the intraclass data and the separability of the interclass data.

For the running time of DR, EHGDL takes more running time than the other methods except SLGDA, because EHGDL needs to construct more models including the intraclass and interclass hypergraphs, the supervised locality graph, and the weighted neighborhood margin. SLGDA takes too much time to solve the sparse coefficients with the alternating direction method of multipliers. For the running time of classification, all the method has a similar time for using the same classifier.

As shown in Figs. 9–11, the proposed method achieves smoother classification maps than the other compared DR algorithms on the three data sets and contains more homogeneous areas, especially, in the regions of Alfalfa, Corn-min, Corn, Soybean-notill, Soybean-min, and Soybean-clean for the Indian Pines data set, Asphalt, Soil, Bitumen, and Bricks for the PaviaU data set, Healthy Grass, Residential, Road, Highway, Railway, and Parking Lot 1 for the HoustonU data set.

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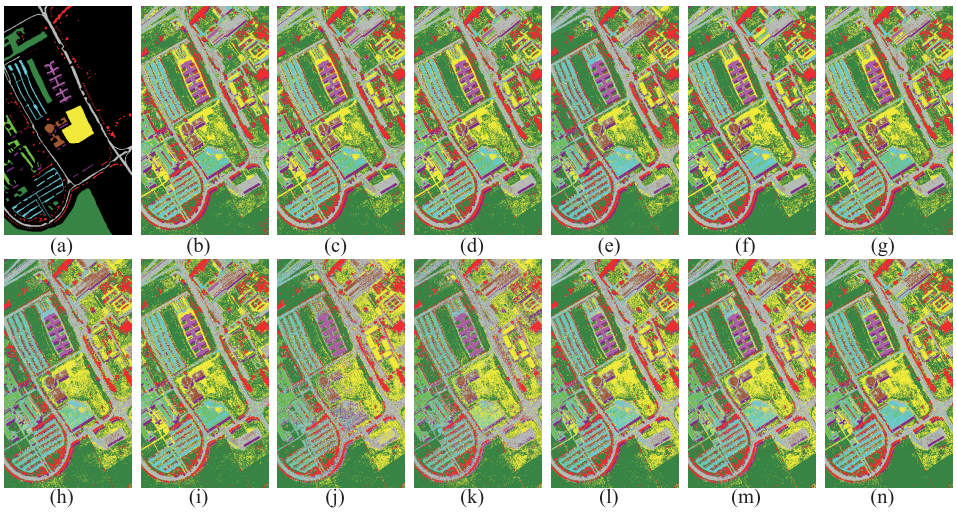



Fig. 10. Classification maps of different methods on the PaviaU data set. (a) Groundtruth. (b) RAW. (c) MMC. (d) LFDA. (e) LGSFA. (f) RLDE. (g) BH. (h) DHLP. (i) SDE. (j) SGDA. (k) CGDA. (l) LGDA. (m) SLGDA. (n) EHGDL.

TABLE IV
CLASSIFICATION ACCURACIES OF EACH CLASS (%) AND RUNNING TIME (SECOND) FOR DIFFERENT METHODS ON THE HOUSTONU DATA SET

| Class | RAW | MMC [34] | LFDA [41] | LGSFA [43] | RLDE [42] | BH [57] | DHLP [56] | SDE [51] | SGDA [47] | CGDA [53] | LGDA [52] | SLGDA [52] | EHGDL |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 97.44 | 97.36 | 97.60 | 98.00 | 97.28 | 97.44 | 97.36 | 97.52 | 96.80 | 97.52 | 97.20 | 97.28 | **98.24** |
| 2 | 97.77 | 97.77 | 98.33 | 98.41 | 99.12 | 98.64 | 97.77 | 98.72 | 98.80 | 99.20 | 99.20 | 99.20 | **99.36** |
| 3 | 97.99 | 97.99 | 98.71 | **100** | **100** | 99.71 | **100** | 98.85 | **100** | **100** | 99.71 | 99.71 | **100** |
| 4 | 98.71 | 98.71 | 98.87 | 98.55 | **99.28** | 98.07 | 98.63 | 98.55 | 96.38 | 97.51 | 97.59 | 97.59 | 98.07 |
| 5 | 98.23 | 98.15 | 98.55 | 99.36 | **99.68** | 99.03 | 98.87 | 98.87 | 98.79 | 98.87 | 98.95 | 99.03 | 99.44 |
| 6 | 94.46 | 94.46 | 95.69 | 99.38 | 99.38 | 98.77 | **99.69** | 96.31 | 97.54 | 97.54 | 97.23 | 97.23 | 99.08 |
| 7 | 83.20 | 83.20 | 87.15 | 90.22 | 92.27 | 91.25 | 88.01 | 89.20 | 93.69 | 93.22 | 94.01 | 94.40 | **95.50** |
| 8 | 85.05 | 85.05 | 87.14 | 90.11 | 90.03 | 89.95 | 88.67 | 88.91 | **93.01** | 92.36 | 90.43 | 90.51 | 92.93 |
| 9 | 78.75 | 78.75 | 83.71 | 87.14 | 90.26 | 90.65 | 74.36 | 88.66 | 84.74 | 90.18 | 90.97 | 92.25 | **96.09** |
| 10 | 86.15 | 86.15 | 89.24 | 91.69 | 92.01 | 92.67 | 87.29 | 89.65 | 95.68 | 96.41 | 93.72 | 93.97 | **98.70** |
| 11 | 85.67 | 85.67 | 90.04 | 92.71 | 86.56 | 94.66 | 83.00 | 90.12 | 93.04 | 93.68 | 93.85 | 94.49 | **97.00** |
| 12 | 81.91 | 81.83 | 85.81 | 88.00 | 84.27 | 88.65 | 77.94 | 89.38 | 92.54 | 95.54 | 91.65 | 92.21 | **96.59** |
| 13 | 33.90 | 33.90 | 35.61 | 57.14 | 55.22 | 35.39 | **59.06** | 40.51 | 49.04 | 48.19 | 51.17 | 51.81 | **59.06** |
| 14 | 98.36 | 98.13 | 98.60 | 99.53 | 99.53 | 98.13 | 99.53 | 98.36 | **99.77** | **99.77** | 99.53 | 99.53 | 99.53 |
| 15 | 99.39 | 99.39 | 99.85 | **100** | **100** | **100** | **100** | 99.85 | **100** | **100** | 99.70 | 99.70 | **100** |
| AA | 87.80 | 87.77 | 89.66 | 92.68 | 92.33 | 91.53 | 90.01 | 90.90 | 92.65 | 93.33 | 92.99 | 93.26 | **95.31** |
| OA | 88.78 | 88.76 | 90.87 | 93.19 | 92.85 | 93.01 | 89.76 | 92.13 | 93.67 | 94.55 | 94.04 | 94.34 | **96.36** |
| KC | 87.86 | 87.83 | 90.12 | 92.63 | 92.27 | 92.43 | 88.93 | 91.48 | 93.15 | 94.10 | 93.55 | 93.87 | **96.06** |
| T1(s) | - | 0.74 | 1.47 | 11.56 | 9.30 | 4.36 | 12.79 | 4.88 | 4.63 | 1.65 | 26.16 | 2620.17 | 94.14 |
| T2(s) | 962.02 | 207.57 | 231.23 | 206.20 | 205.67 | 211.29 | 213.70 | 220.59 | 232.19 | 299.29 | 230.94 | 230.44 | 293.80 |

Because the proposed method considers the complementarity of different information of HSI which can better reveal the intrinsic properties of HSI and obtain stronger discriminant features for classification.

### *E. Analysis for Parameters*

For the EHGDL method, it contains two trade-off parameters, two neighbor variables, and an intraclass weighted parameter. It is very important to select proper parameters for better classification results. Therefore, we analyze the change of OAs with different parameter values in this section.

*1) Tradeoff Parameters $\alpha$ and $\beta$:* The trade-off parameters $\alpha$ and $\beta$ are used to control the contribution of the supervised locality graph and the weighted neighborhood margin. We repeated the experiments with 10 times in each condition on the three data sets, and the average OAs with different values of $\alpha$ and $\beta$ were given in Fig. 12.

According to Fig. 12, on the three data sets, the OAs both improve with the increasing of $\alpha$ and $\beta$, and then the results begin to decline when $\alpha$ and $\beta$ exceed a certain value because a large value will reduce the contribution of the two discriminant hypergraphs while a small value will weaken the contribution of the supervised locality graph and the weighted neighborhood margin. If $\alpha$, $\beta$, or both of them are set to zero, EHGDL cannot obtain the best classification accuracies and the proposed model will degenerate to HGDL, EHDL, and HDL, which indicates that the supervised locality model and the weighted neighborhood margin are benefit to enhance the discriminant performance of EHGDL. Therefore, we obtain a series of optimal results for $\alpha$ and $\beta$ in a certain range, and $\alpha$ and $\beta$ are set to 0.8 and 1.6 for the Indian Pines data set, 3.2 and 2.2 for the PaviaU data set, 3 and 1 for the HoustonU data set in the experiments.

*2) Neighbor Size $k_w$ and $k_b$:* For the proposed method, we adopt different ways to select the $k_w$ intraclass neighbors and the $k_b$ interclass neighbors for the construction of the discriminant hypergraphs and the weighted neighborhood margin, which can reveal different neighbor information to represent

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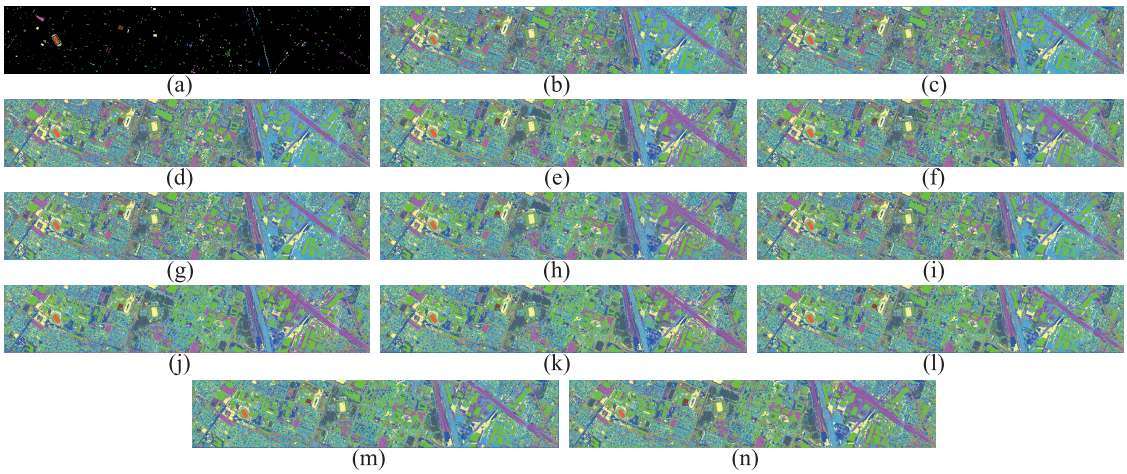



Fig. 11. Classification maps of different methods on the HoustonU data set. (a) Groundtruth. (b) RAW. (c) MMC. (d) LFDA. (e) LGSFA. (f) RLDE. (g) BH. (h) DHLP. (i) SDE. (j) SGDA. (k) CGDA. (l) LGDA. (m) SLGDA. (n) EHGDL.

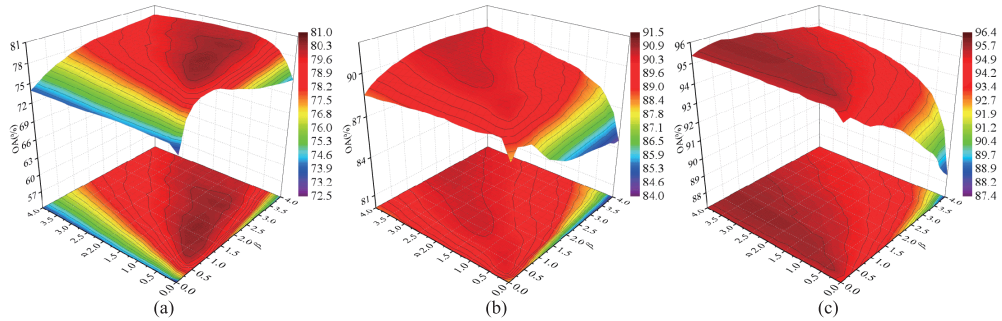



Fig. 12. OAs with respect to the tradeoff parameters $\alpha$ and $\beta$ on (a) Indian Pines, (b) PaviaU, and (c) HoustonU data sets.

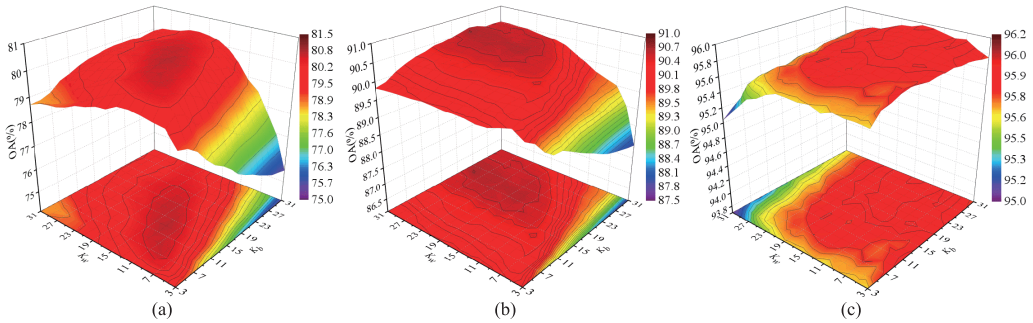



Fig. 13. OAs with respect to the neighbor size $k_w$ and $k_b$ on (a) Indian Pines, (b) PaviaU, and (c) HoustonU data sets.

different intrinsic structures. The OAs with different $k_w$ and $k_b$ are shown in Fig. 13.

From Fig. 13, we can draw some similar conclusions on the three data sets. With the increase of $k_w$ and $k_b$, the OAs first increase to a peak value and then decrease when $k_w$ and $k_b$ exceed a certain value. For the intraclass neighbors, a large $k_w$ will contain some data with a large difference to influence the compactness of the intraclass data, while there is insufficient information to represent the intrinsic properties of the intraclass data with a small $k_w$. For the interclass neighbors, we aim to separate similar samples from different classes. Some similar samples from different classes will not be selected with a small $k_b$, whereas a large $k_b$ will introduce some additional information to affect the construction of the training model. Therefore, some proper values of $k_w$ and $k_b$ should be used in the experiments to obtain some desired classification results, and we set $k_w$ and $k_b$ to 13 and 13 for the Indian Pines data set, 19 and 21 for the PaviaU data set, 15 and 23 for the HoustonU data set.

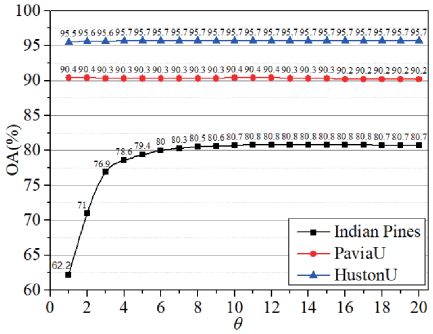



Fig. 14. OAs with respect to the intraclass weighted parameter $\theta$ on the three data sets.

*3) Intraclass Weighted Parameter $\theta$:* In addition, we used a weight $\theta$ to enhance the contribution of the intraclass data

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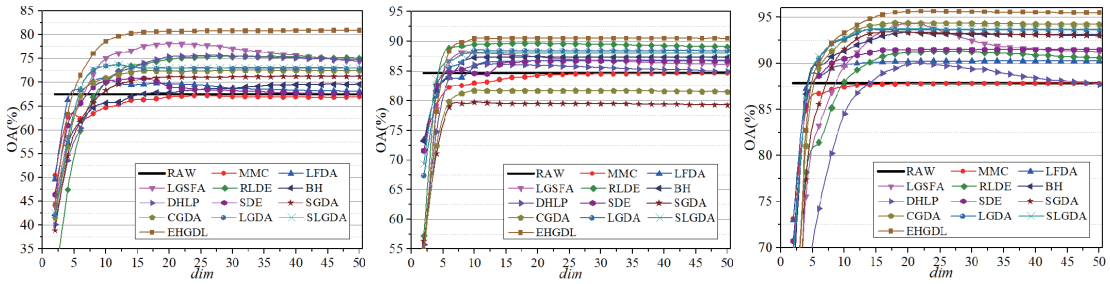



Fig. 15. OAs with respect to different embedding dimensions on (a) Indian Pines, (b) PaviaU, and (c) HoustonU data sets.

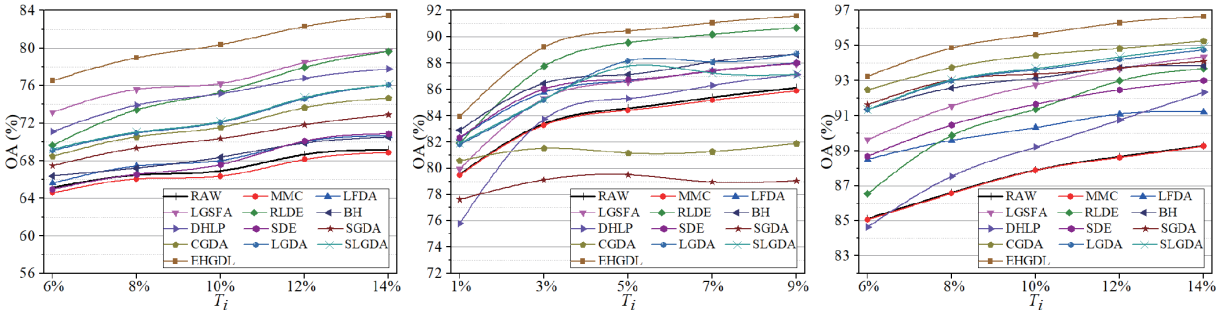



Fig. 16. OAs with respect to different numbers of training samples on (a) Indian Pines, (b) PaviaU, and (c) HoustonU data sets.

TABLE V

CLASSIFICATION RESULTS WITH DIFFERENT CLASSIFIERS ON THE THREE DATA SETS (OA ± STD (KC) %)

| Method | Indian Pines | | | PaviaU | | | HoustonU | | |
|---|---|---|---|---|---|---|---|---|---|
| | NN | SAM | MLC | NN | SAM | MLC | NN | SAM | MLC |
| RAW | 67.73±0.46 (63.16) | 68.82±0.44 (64.39) | 69.29±0.89 (64.44) | 84.62±0.31 (79.38) | 83.45±0.45 (77.79) | 71.24±0.64 (58.74) | 87.63±0.37 (86.61) | 91.13±0.26 (90.40) | 76.46±2.83 (74.51) |
| MMC[34] | 67.10±0.65 (62.44) | 68.33±0.58 (63.83) | 60.13±2.09 (54.72) | 84.44±0.48 (79.15) | 83.27±0.70 (77.56) | 89.40±0.49 (85.80) | 87.58±0.39 (86.56) | 91.00±0.25 (90.26) | 91.88±0.33 (91.21) |
| LFDA[41] | 69.17±0.55 (64.78) | 69.03±0.49 (64.61) | 60.35±1.67 (54.75) | 86.73±0.29 (82.22) | 84.70±0.28 (79.47) | 89.41±0.37 (85.82) | 90.17±0.31 (89.37) | 91.83±0.24 (91.17) | 91.17±0.34 (90.45) |
| LGSFA[43] | 77.50±0.50 (74.22) | 77.33±0.51 (74.07) | 75.34±1.17 (71.83) | 86.96±0.50 (82.37) | 83.46±0.69 (77.85) | 85.32±0.38 (80.31) | 92.59±0.40 (91.98) | 92.74±0.40 (92.15) | 90.39±0.34 (89.61) |
| RLDE[42] | 76.26±0.90 (72.75) | 76.06±1.00 (72.53) | 73.57±2.13 (69.97) | 89.50±0.39 (85.90) | 85.36±1.12 (80.27) | 90.16±0.46 (86.84) | 91.68±0.46 (91.00) | 90.80±0.54 (90.05) | 88.30±0.36 (87.34) |
| BH[57] | 68.93±0.42 (64.48) | 68.90±0.47 (64.45) | 63.18±2.66 (58.03) | 87.76±0.56 (83.62) | 86.33±0.47 (81.66) | 89.19±0.30 (85.53) | 92.92±0.39 (92.34) | 92.64±0.40 (92.04) | **92.37**±0.28 (91.75) |
| DHLP[56] | 75.90±0.72 (72.40) | 76.28±0.43 (72.84) | 74.64±1.59 (71.01) | 85.62±0.62 (80.83) | 84.50±0.45 (79.32) | 85.89±0.29 (81.12) | 89.58±0.42 (88.73) | 89.15±0.44 (88.27) | 87.07±0.36 (86.01) |
| SDE[51] | 68.51±0.63 (64.00) | 68.69±0.73 (64.18) | 64.40±1.78 (59.08) | 86.63±0.49 (82.05) | 83.67±0.45 (78.06) | 89.49±0.37 (85.93) | 91.50±0.27 (90.80) | 93.38±0.32 (92.84) | 92.14±0.25 (91.49) |
| SGDA[47] | 70.89±0.59 (66.66) | 68.03±0.62 (63.47) | 68.69±0.89 (63.86) | 79.45±0.48 (72.66) | 71.60±0.59 (61.50) | 79.57±0.53 (72.63) | 93.26±0.34 (92.71) | 92.10±0.47 (91.45) | 89.66±0.38 (88.81) |
| CGDA[53] | 72.21±0.67 (68.20) | 70.45±0.55 (66.22) | 70.14±0.95 (65.58) | 81.43±0.70 (75.26) | 72.11±0.83 (62.15) | 82.57±0.63 (76.54) | 94.22±0.27 (93.75) | 93.33±0.39 (92.78) | 90.52±0.29 (89.74) |
| LGDA[52] | 73.55±1.03 (69.77) | 72.16±0.90 (68.11) | 69.43±0.99 (64.89) | 88.20±0.41 (84.27) | 84.39±0.46 (79.06) | 88.17±0.24 (84.24) | 93.50±0.42 (92.97) | 93.25±0.31 (92.70) | 90.54±0.37 (89.77) |
| SLGDA[52] | 73.57±1.02 (69.79) | 72.17±0.90 (68.12) | 69.44±0.97 (64.91) | 87.65±0.35 (83.54) | 82.43±0.47 (76.38) | 87.03±0.45 (82.64) | 93.60±0.42 (93.08) | 93.32±0.32 (92.77) | 90.55±0.40 (89.78) |
| EHGDL | **80.81**±1.00 (78.05) | **81.05**±0.70 (78.33) | **77.59**±1.42 (74.30) | **90.45**±0.33 (87.22) | **88.57**±0.29 (84.66) | **90.60**±0.27 (87.43) | **95.57**±0.59 (95.21) | **95.74**±0.63 (95.39) | 92.36±0.53 (91.74) |

in the construction of the supervised locality graph, and we analyzed the classification accuracies with different $\theta$ as shown in Fig. 14.

With Fig. 14, for the Indian Pines data set, the classification accuracies improve with the increase of $\theta$, and then tend to maintain a stable peak value when the value of $\theta$ continues to rise because a large $\theta$ will enhance the contribution of the intraclass data. When $\theta$ exceeds a certain value, the contribution of the intraclass samples will achieve the best effect and generate a stable classification accuracy. For the PaviaU and HoustonU data sets, we can obtain the OAs with a subtle change when the value of $\theta$ increases, which explains that the value of $\theta$ has a small impact on the classification results. Therefore, we select a proper value of $\theta$ in the experiments and set it to 15 for the Indian Pines data set, 1 for the PaviaU data set, and 9 for the HoustonU data set.

### *F. Embedding Dimension*

For a DR method, the embedding dimension will affect the classification accuracy, so we analyze the change of OAs with

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respect to different embedding dimensions (*dim*) as shown in Fig. 15.

According to Fig. 15, the OAs quickly ascend with the increase of the embedding dimension and then reach a stable peak value for the most algorithms, since large embedding features will possess more information and better discriminant performance. When the features reach the maximum discriminating power, the classification accuracies will keep a certain value in spite of continuously increasing the dimensionality. In addition, some methods result in the Hughes phenomenon that the OAs first rise and then decline with the increase of the embedding dimension, i.e., LGSFA and SDE for the Indian Pines data set, LGSFA and DHLP for the HoustonU data set. Thus, for obtaining better results, we set the embedding dimension to 25 for all the methods on the three data sets. In Fig. 15, most methods can generate better results than RAW because these methods can reduce the redundant information of data and enhance the discriminant performance of the embedding features. For all the methods, the proposed EHGDL algorithm obtains the best classification accuracies in most cases compared with the other state-of-the-art methods.

### G. Sample Size

To demonstrate the influence under different numbers of training samples, we selected $T_i$ samples from each class as different training sets, where $T_i$ is set to {6%, 8%, 10%, 12%, 14%} for the Indian Pines and HoustonU data sets, {1%, 3%, 5%, 7%, 9%} for the PaviaU data set. The classification results are displayed in Fig. 16.

According to Fig. 16, for all the methods, the OAs improve with the increase of the number of training samples on the three data sets, thus indicating that more training samples possess more desired information to represent the embedding features. The proposed method achieves the best OAs under different numbers of training samples, which indicates that EHGDL can effectively represent the complex intrinsic properties of HSI with the two discriminant hypergraphs and the supervised locality graph, and it also enhances the compactness of the intraclass samples and the separability of the interclass samples with the weighted neighborhood margin, which improves the discriminating power of the low-dimensional embedding features.

### H. Results With Different Classifiers

Since different classifiers will generate different classification results, to analyze the effectiveness of the DR algorithms with different classifiers, we selected the NN, SAM, and MLC methods to perform the classification after extracting the embedding features. Table V shows the average OAs with STD and the average KCs under different classifiers.

In Table V, with the NN, SAM, and MLC classification models, EHGDL achieves better OAs and KCs than the other DR methods, because the proposed method considers different intrinsic relationships, including the intraclass hypergraph, the interclass hypergraph, the supervised locality graph, and the weighted neighborhood margin, to obtain the low-dimensional features of HSI. The classification results indicate that the extracted features from the EHGDL method possess better robustness for different classifiers.

## V. Conclusion

In this article, we proposed a DR method, termed EHGDL, based on the theories of graph, hypergraph, and margin to obtain the discriminant features of the HSI for the classification of land-cover. In EHGDL, we design an intraclass hypergraph and an interclass hypergraph with the intraclass and interclass neighbors, respectively. To achieve more intrinsic information, we construct a supervised locality graph with the class label and the locality-constrained coding. To further enhance the discriminant performance, we construct a weighted neighborhood margin using the KLD. According to the hypergraph, graph, and margin models, a DR model is constructed to compact the intraclass samples by the minimization of the intraclass hypergraph and the supervised locality graph and separate the interclass samples by the maximization of the interclass hypergraph and the weighted neighborhood margin, and then a projection matrix can be obtained with this DR model. Finally, we can extract the discriminant features of the HSI with the projection matrix. EHGDL considers not only the multiple relationships but also the binary relationships of data, as well as the marginal information to reveal the features of HSI, which can achieve the complementarity of different properties of HSI. Experiments on the Indian Pines, PaviaU, and HoustonU data sets show that EHGDL achieve better classification accuracies compared with the traditional graph or hypergraph learning methods because the proposed method combines the merits of graph and hypergraph to form the complementarity of different information. As a result, EHGDL can effectively reveal the complex intrinsic relationships of HSI and obtain strong discriminant features to improve the classification of HSI. However, our method takes too much time to design the hypergraph, graph, and margin models, so we will focus on how to reduce the running time and rise the computational efficiency in the future work. In addition, we will use adaptive relationship representation instead of KNNs to effectively reveal the high-order structure of data.

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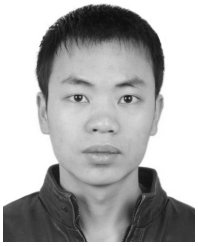
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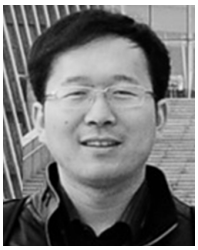
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